



**Zagazig University**

**Faculty of Engineering**



# **ECE 321: Electronic Circuits**

**Lecture (03) & (04) → Amplifiers & Current Mirrors**

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**ECE 321 Electronic Circuits, ECE dept., Faculty of Engineering, Zagazig University, Fall 2022,  
by Dr. Ahmed Wahba**

# Outline

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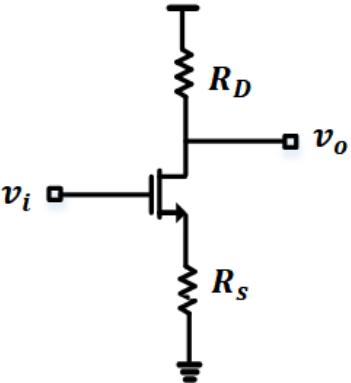
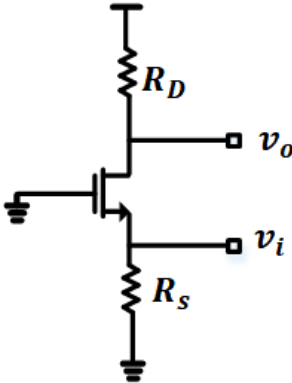
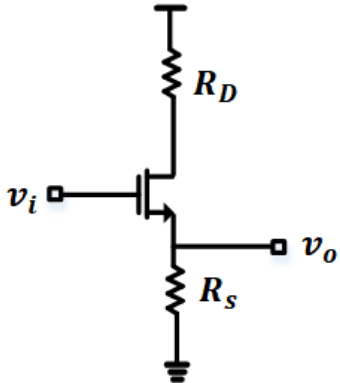
- ❑ Amplifiers (Get the gain using  $G_m \times R_{out}$ )
- ❑ Current Mirrors
  - ❑ Ideal Current Source
  - ❑ Transistor as Current Source
  - ❑ Current Mirror Specifications
  - ❑ MOSFET Current Mirrors
  - ❑ BJT Current Mirrors

# Outline

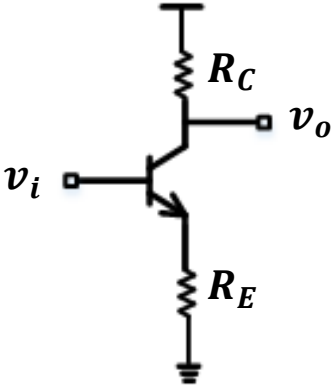
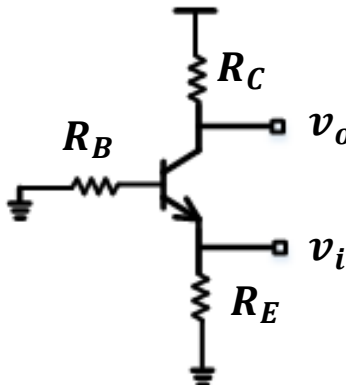
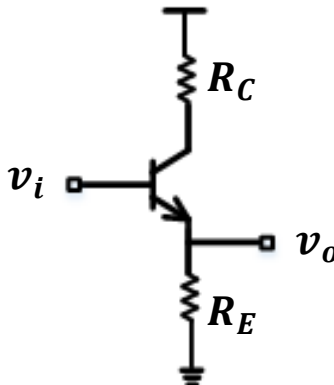
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- ❑ **Amplifiers (Get the gain using  $G_m \times R_{out}$ )**
- ❑ **Current Mirrors**
  - ❑ **Ideal Current Source**
  - ❑ **Transistor as Current Source**
  - ❑ **Current Mirror Specifications**
  - ❑ **MOSFET Current Mirrors**
  - ❑ **BJT Current Mirrors**

# Summary of Basic MOSFET Amplifier Topologies

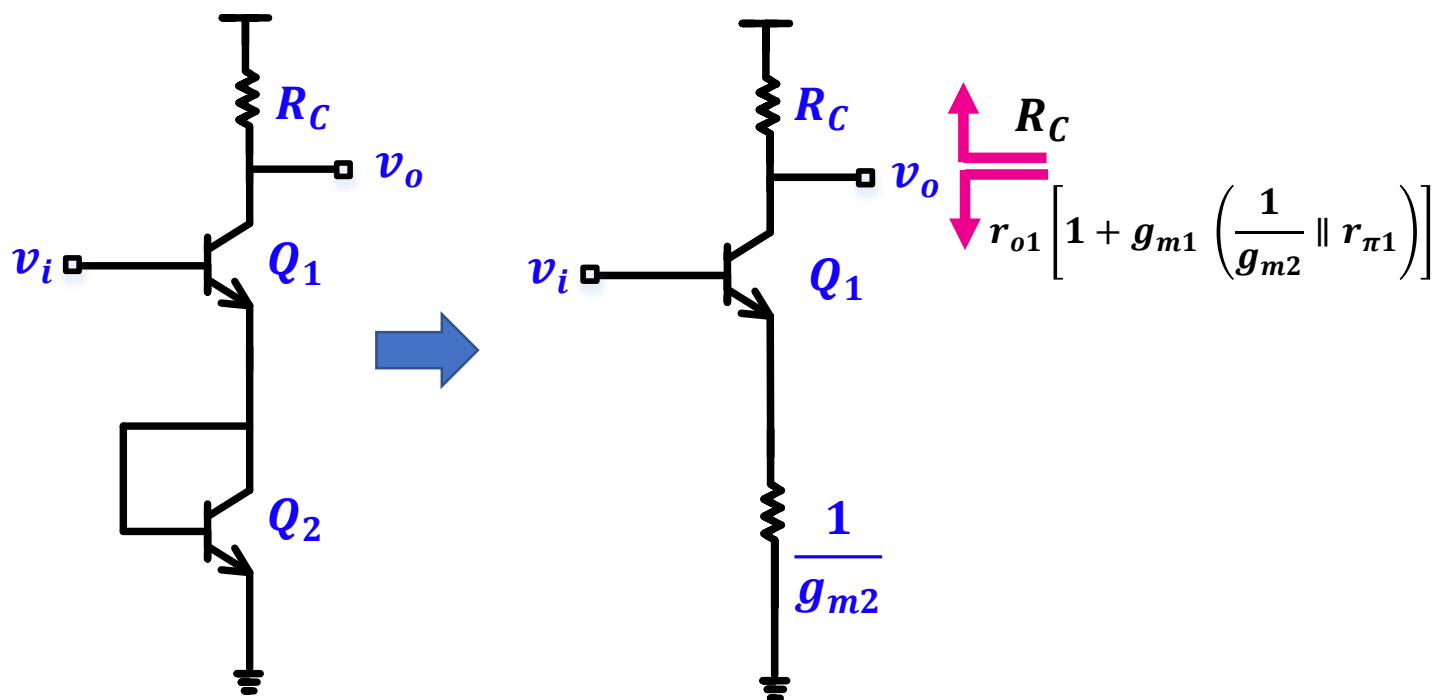
|           | Common Source                                                                      | Common Gate                                                                         | Common Drain                                                                        |
|-----------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|           |  |  |  |
| $R_{in}$  | $\infty$                                                                           | $R_S \parallel \frac{1}{g_m}$                                                       | $\infty$                                                                            |
| $R_{out}$ | $R_D$                                                                              | $R_D$                                                                               | $R_S \parallel \frac{1}{g_m} = \frac{R_S}{1 + g_m R_S}$                             |
| $G_m$     | $\frac{-g_m}{1 + g_m R_S}$                                                         | $g_m$                                                                               | $g_m$                                                                               |
| $A_v$     | $\frac{-g_m R_D}{1 + g_m R_S}$                                                     | $g_m R_D$                                                                           | $\frac{g_m R_S}{1 + g_m R_S}$                                                       |

# Summary of Basic BJT Amplifier Topologies

|           | Common Emitter                                                                    | Common Base                                                                                                                      | Common Collector                                                                    |
|-----------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|           |  |                                               |  |
| $R_{in}$  | $r_{\pi}[1 + g_m R_E]$                                                            | $R_E \parallel \frac{1}{g_m} \left[ 1 + \frac{R_B}{r_{\pi}} \right] \approx R_E \parallel \frac{1}{g_m}$<br>if $r_{\pi} \gg R_B$ | $r_{\pi}[1 + g_m R_E]$                                                              |
| $R_{out}$ | $R_C$                                                                             | $R_C$                                                                                                                            | $R_E \parallel \frac{1}{g_m} = \frac{R_E}{1 + g_m R_E}$                             |
| $G_m$     | $\frac{-g_m}{1 + g_m R_E}$                                                        | $\frac{g_m}{1 + \frac{R_B}{r_{\pi}}} \approx g_m$<br>if $r_{\pi} \gg R_B$                                                        | $g_m$                                                                               |
| $A_v$     | $\frac{-g_m R_C}{1 + g_m R_E}$                                                    | $\frac{g_m R_C}{1 + \frac{R_B}{r_{\pi}}} \approx g_m R_C$<br>if $r_{\pi} \gg R_B$                                                | $\frac{g_m R_E}{1 + g_m R_E}$                                                       |

# Example # 1

□ Find  $R_{in}$ ,  $R_{out}$ , and  $A_v$ . Note: All transistors operate in active region and detailed biasing is not shown. Output resistor of  $Q_n$  is  $r_{on}$ .



B & C of  $Q_2$  are shorted  $\rightarrow$  replace  $Q_1$  with  $1/g_{m2}$

$$\begin{array}{c} r_o [1 + g_m (R_E \parallel r_\pi)] \\ \swarrow \searrow \\ r_\pi [1 + g_m R_E] \quad \frac{1}{g_m} \left[ 1 + \frac{R_B}{r_\pi} \right] \end{array}$$

$$\square R_{in} = r_{\pi 1} \left[ 1 + \frac{g_{m1}}{g_{m2}} \right]$$

$$\square R_{out} = R_C$$

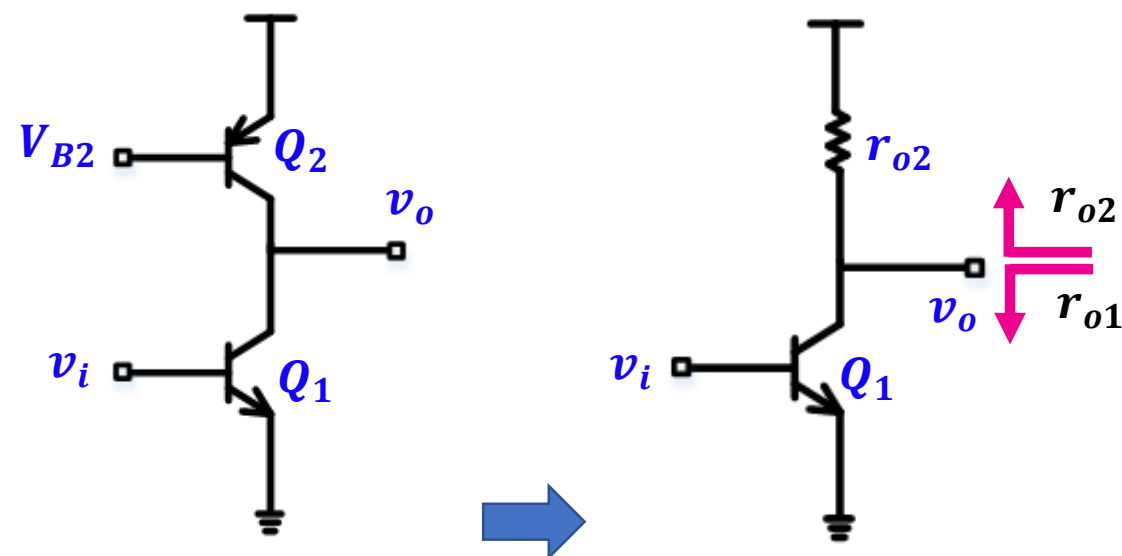
$$\square G_m = \frac{-g_{m1}}{1 + \frac{g_{m1}}{g_{m2}}}$$

$$\square A_v = G_m \times R_{out} = \frac{-g_{m1} R_C}{1 + \frac{g_{m1}}{g_{m2}}} = \frac{-g_{m1} g_{m2} R_C}{g_{m1} + g_{m2}}$$

# Example # 2

- Find  $R_{in}$ ,  $R_{out}$ , and  $A_v$ . Note: All transistors operate in active region and detailed biasing is not shown. Output resistor of  $Q_n$  is  $r_{on}$ .

$$\begin{array}{c} r_o[1 + g_m(R_E \parallel r_\pi)] \\ \swarrow \searrow \\ r_\pi[1 + g_m R_E] \quad \frac{1}{g_m} \left[ 1 + \frac{R_B}{r_\pi} \right] \end{array}$$



$V_{B2}$  is a DC voltage  $\rightarrow$   
replaced with AC ground

B and E of  $Q_2$  are AC ground  
 $\rightarrow Q_2$  is replaced by  $r_{o2}$

Note:  $Q_2$  is called Active load.

□  $R_{in} = r_{\pi 1}$

□  $R_{out} = r_{o1} \parallel r_{o2}$

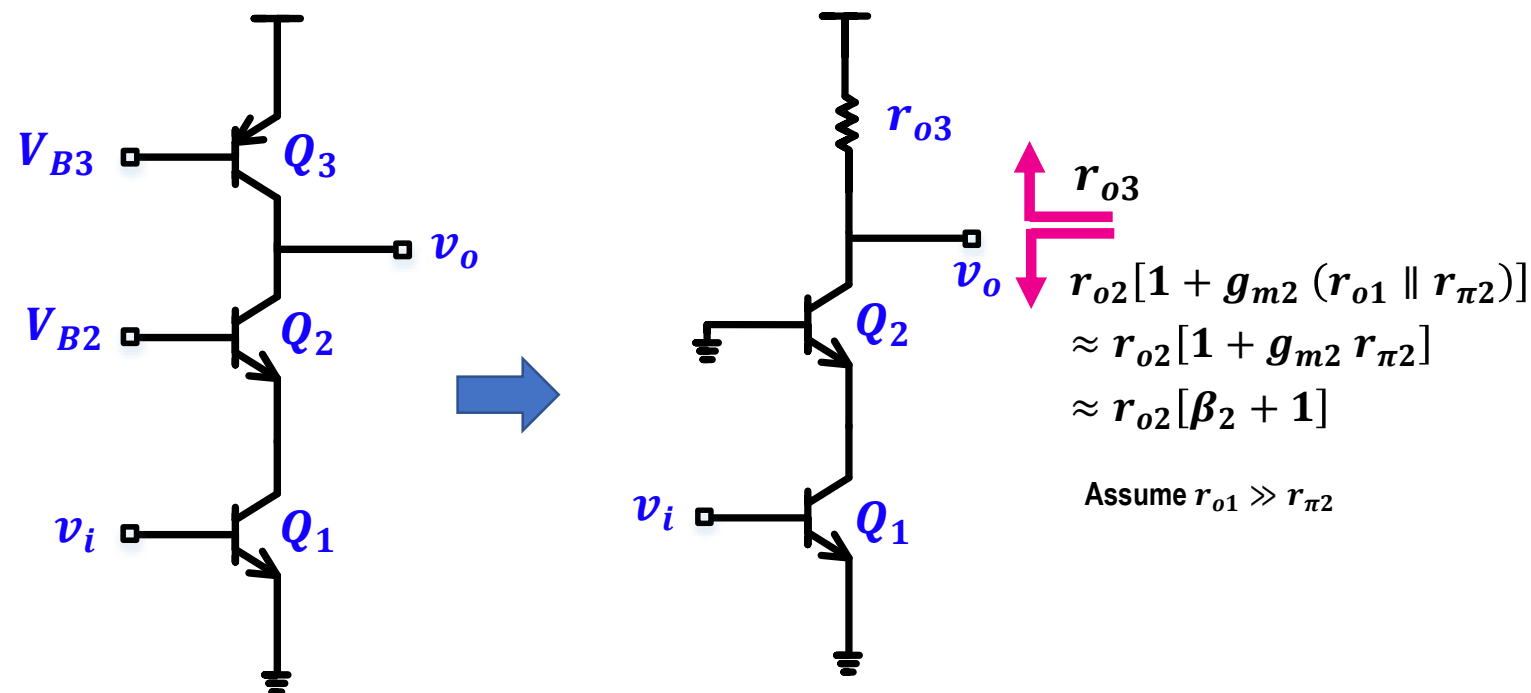
□  $G_m = -g_m$

□  $A_v = G_m \times R_{out} = -g_{m1}(r_{o1} \parallel r_{o2})$

# Example # 3

□ Find  $R_{in}$ ,  $R_{out}$ , and  $A_v$ . Note: All transistors operate in active region and detailed biasing is not shown. Output resistor of  $Q_n$  is  $r_{on}$ .

$$\begin{array}{c} r_o[1 + g_m(R_E \parallel r_\pi)] \\ \swarrow \searrow \\ r_\pi[1 + g_m R_E] \quad \frac{1}{g_m} \left[ 1 + \frac{R_B}{r_\pi} \right] \end{array}$$



$V_{B2}$  and  $V_{B3}$  are DC voltage  
→ replaced with AC ground

B and E of  $Q_3$  are AC ground  
→  $Q_3$  is replaced by  $r_{o3}$

Note: this structure is called cascode amplifier (to be discussed in detail later)

$$\square R_{in} = r_{\pi 1}$$

$$\square R_{out} = r_{o3}$$

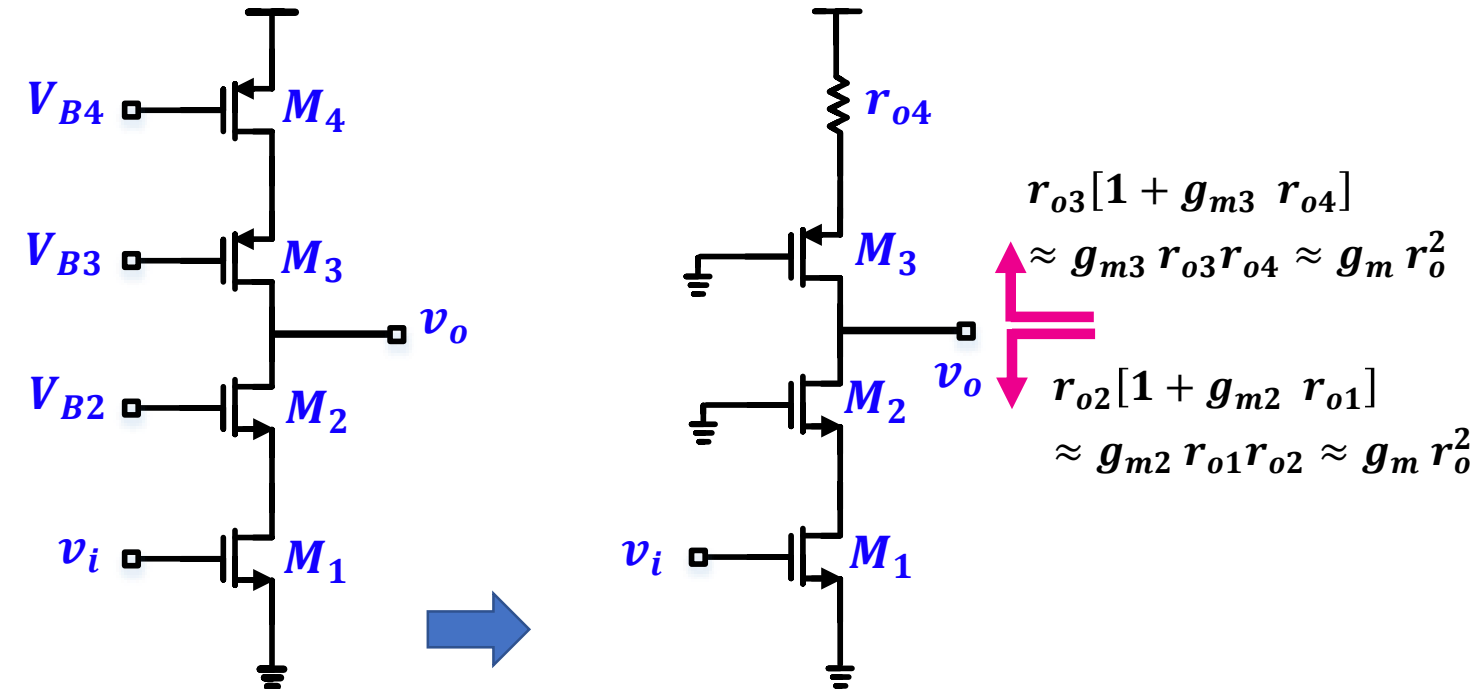
$$\square G_m = -g_m$$

$$\square A_v = G_m \times R_{out} = -g_{m1} r_{o3}$$

# Example # 4

- Find  $R_{in}$ ,  $R_{out}$ , and  $A_v$ . Note: All transistors operate in saturation region and detailed biasing is not shown. Assume that all transistors have the same  $g_m$  and  $r_o$

$$\infty \rightarrow \begin{matrix} r_o[1 + g_m R_S] \\ \frac{1}{g_m} \left[ 1 + \frac{R_D}{r_o} \right] \end{matrix}$$



$V_{B2}$ ,  $V_{B3}$ , and  $V_{B4}$  are DC voltage  
→ replaced with AC ground

G and S of M4 are AC ground  
→ M4 is replaced by  $r_{o4}$

Note: this structure is called cascode amplifier (to be discussed in detail later)

□  $R_{in} = \infty$

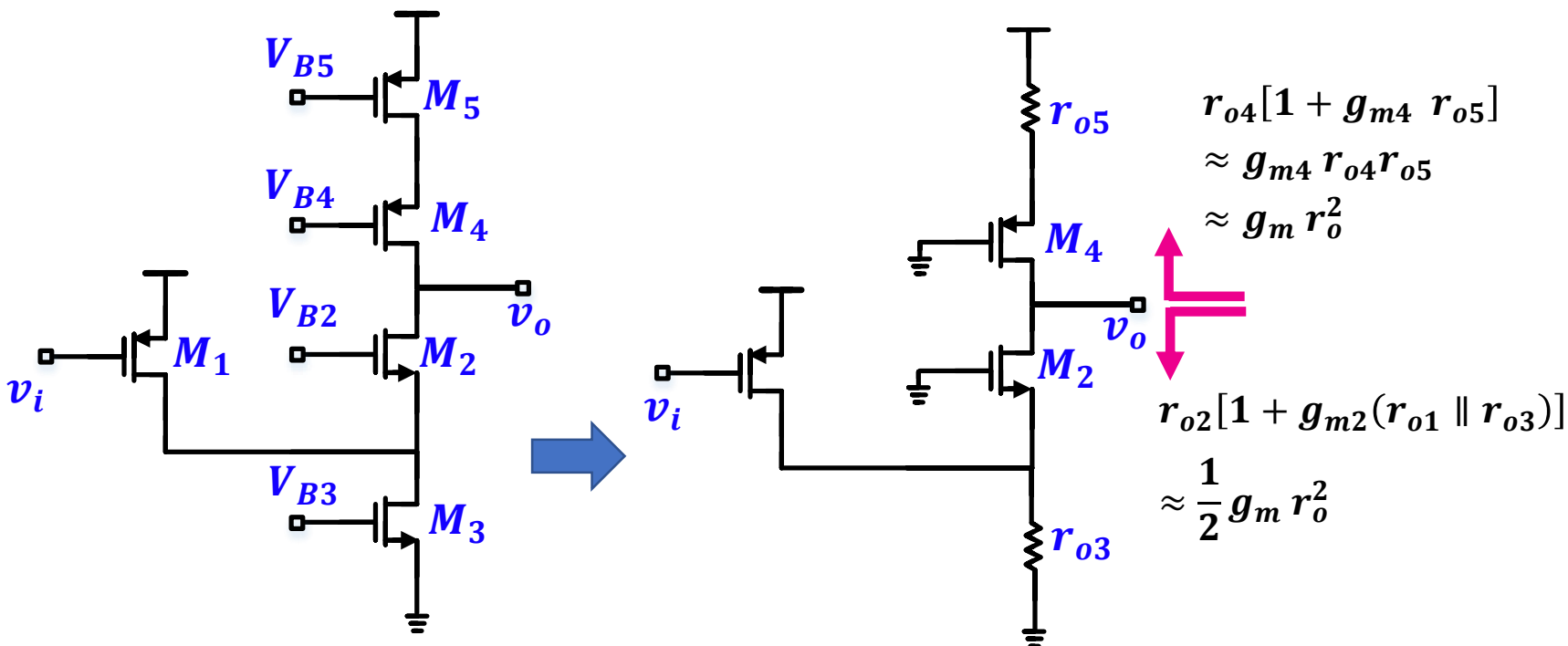
□  $R_{out} \approx g_m r_o^2 \parallel g_m r_o^2 \approx \frac{1}{2} g_m r_o^2$

□  $G_m = -g_{m1} = -g_m$

□  $A_v = G_m \times R_{out} = -\frac{1}{2} (g_m r_o)^2$

# Example # 5

- Find  $R_{in}$ ,  $R_{out}$ , and  $A_v$ . Note: All transistors operate in saturation region and detailed biasing is not shown. Assume that all transistors have the same  $g_m$  and  $r_o$



$V_{Bi}$  is a DC voltage  $\rightarrow$   
replaced with AC ground

M3 and M5 are replaced  
by  $r_{o3}$  and  $r_{o5}$

Note: this structure is called folded cascode amplifier (to be discussed in detail later)

$$\infty \parallel \frac{r_o[1 + g_m R_S]}{\frac{1}{g_m} \left[ 1 + \frac{R_D}{r_o} \right]}$$

□  $R_{in} = \infty$

□  $R_{out} \approx g_m r_o^2 \parallel \frac{1}{2} g_m r_o^2 = \frac{1}{3} g_m r_o^2$

□  $G_m = -g_{m1} = -g_m$

□  $A_v = G_m \times R_{out} = -\frac{1}{3} (g_m r_o)^2$

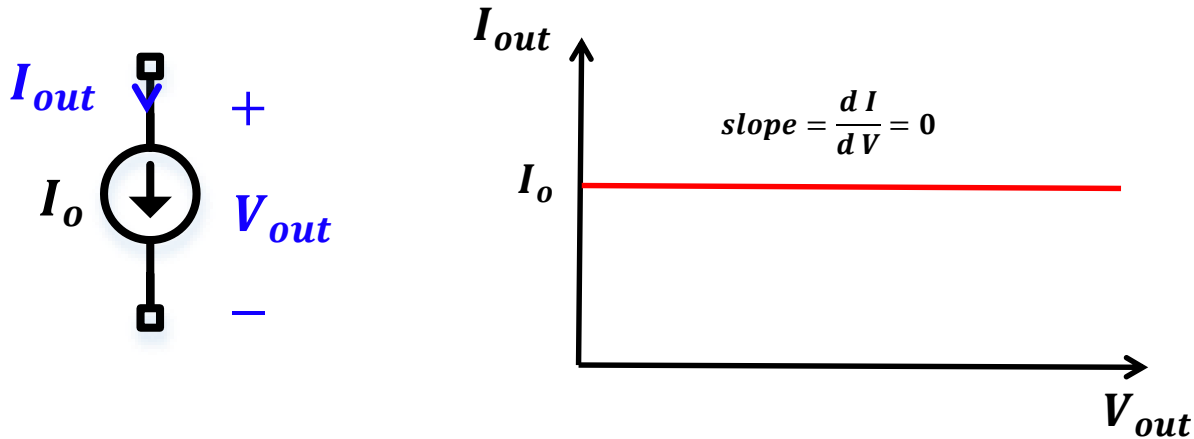
# Outline

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- ❑ Amplifiers (Get the gain using  $G_m \times R_{out}$ )
- ❑ Current Mirrors
  - ❑ Ideal Current Source
  - ❑ Transistor as Current Source
  - ❑ Current Mirror Specifications
  - ❑ MOSFET Current Mirrors
  - ❑ BJT Current Mirrors

# Ideal Current Source

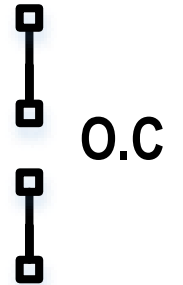
## ☐ Ideal Current Source ( $R_{out} = \infty$ )



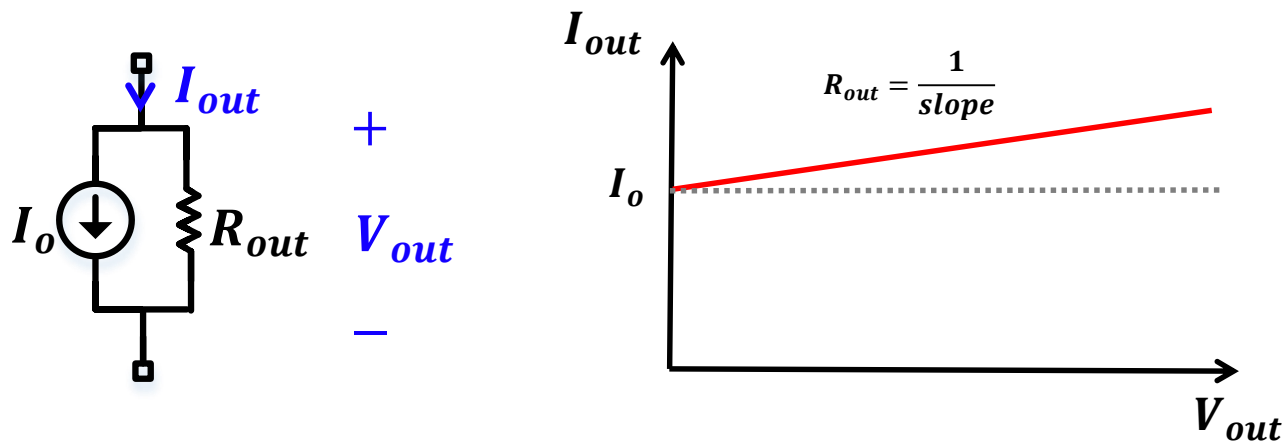
☐  $I_{out} = I_o$  (constant)

☐  $R_{out} = \frac{dV}{dI} = \frac{1}{slope} = \frac{1}{0} = \infty$

## ☐ AC model



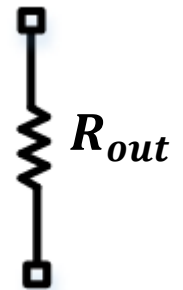
## ☐ Current Source with finite $R_{out}$ ( $R_{out} \neq \infty$ )



☐  $I_{out} = I_o + \frac{V_{out}}{R_{out}}$

☐  $I_{out}$  depends on  $R_{out}$

## ☐ AC model



# Outline

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- ❑ Amplifiers (Get the gain using  $G_m \times R_{out}$ )

- ❑ Current Mirrors

  - ❑ Ideal Current Source

  - ❑ Transistor as Current Source

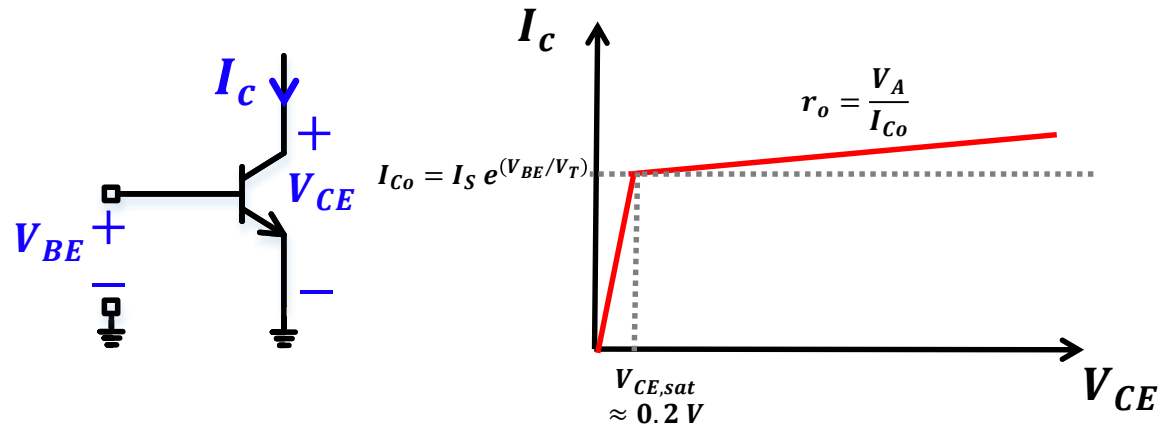
  - ❑ Current Mirror Specifications

  - ❑ MOSFET Current Mirrors

  - ❑ BJT Current Mirrors

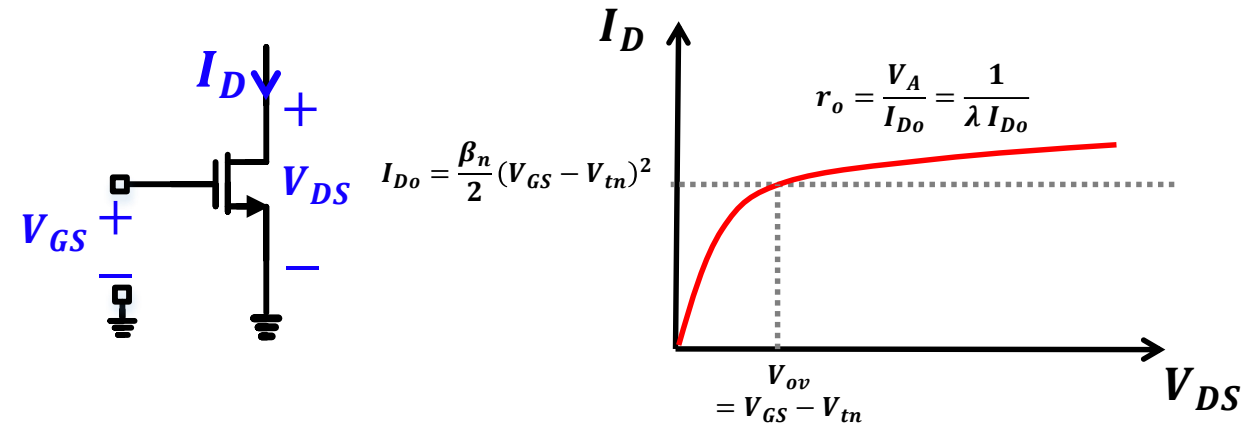
# Transistor as Current Source

## 1- BJT transistor in Active region



- ❑  $I_C = I_S e^{(V_{BE}/V_T)} \left[ 1 + \frac{V_{CE}}{V_A} \right] = I_{C0} \left[ 1 + \frac{V_{CE}}{V_A} \right]$
- ❑  $I_C$  is controlled by  $V_{BE}$
- ❑ For proper operation,  $V_{CE} > V_{CE,sat} (\approx 0.2 \text{ V})$

## 2- MOSFET transistor in Saturation region



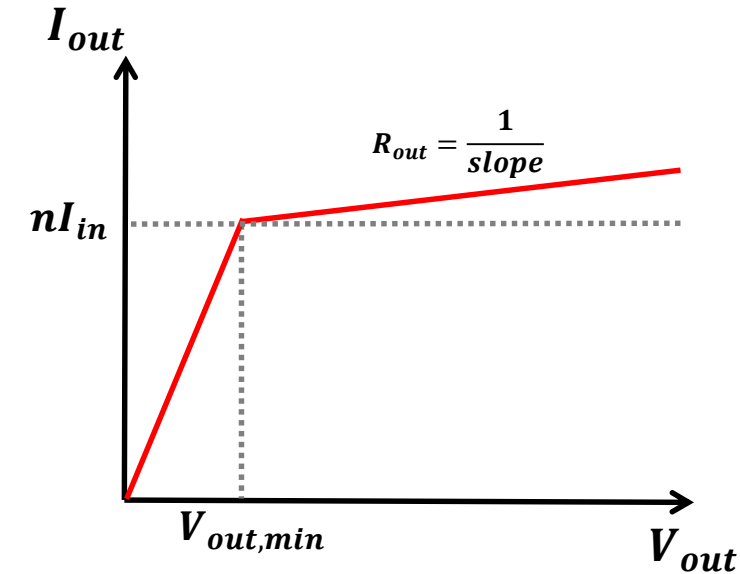
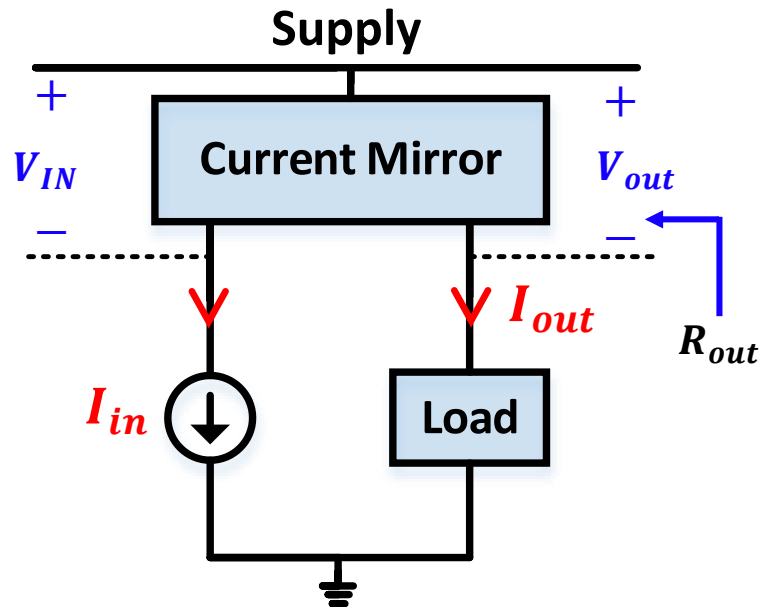
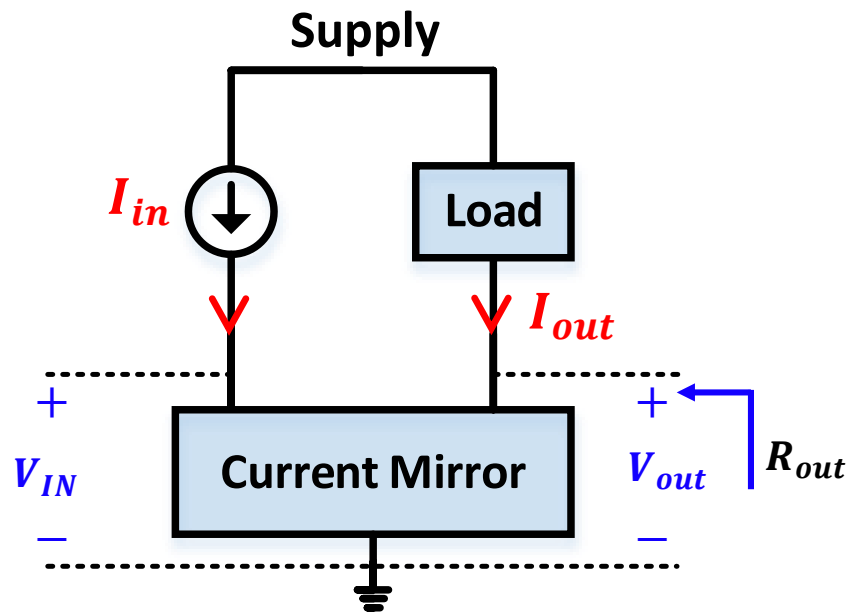
- ❑  $I_D = \frac{\beta_n}{2} (V_{GS} - V_{tn})^2 \times [1 + \lambda V_{DS}] = I_{D0} [1 + \lambda V_{DS}]$
- ❑  $I_D$  is controlled by  $V_{GS}$
- ❑ For proper operation,  $V_{DS} > V_{ov}$

# Outline

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- ❑ Amplifiers (Get the gain using  $G_m \times R_{out}$ )
- ❑ Current Mirrors
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  - ❑ **Current Mirror Specifications**
  - ❑ MOSFET Current Mirrors
  - ❑ BJT Current Mirrors

# Current Mirror Specifications



□  $I_{out} = n \times I_{in}$

□ **Lower  $V_{IN}$**  is required  $\rightarrow$  making the current mirror suitable to operate with lower supply voltage

□ **Lower  $V_{out,min}$**  is required  $\rightarrow$  provide larger headroom for the load.

□ **Large  $R_{out}$**  is required  $\rightarrow I_{out}$  is independent of  $V_{out}$  ( $I_{out} = \text{const}$ ).

# Outline

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- ❑ Current Mirrors
  - ❑ Ideal Current Source
  - ❑ Transistor as Current Source
  - ❑ Current Mirror Specifications
  - ❑ **MOSFET Current Mirrors** (Simple, Cascode, Modified Cascode, High swing, Sootch, Wilson)
  - ❑ BJT Current Mirrors

# Simple MOSFET Current Mirror

□ Get  $\frac{I_{out}}{I_{in}}$

□  $I_G = 0, V_{GS1} = V_{GS2} = V_{GS} \rightarrow V_{ov1} = V_{ov2} = V_{ov}$

□  $I_{in} = I_{D1} = \frac{1}{2} \mu_n C_{ox} (W/L)_1 (V_{GS1} - V_{tn})^2 (1 + \lambda V_{DS1})$

□  $I_{out} = I_{D2} = \frac{1}{2} \mu_n C_{ox} (W/L)_2 (V_{GS2} - V_{tn})^2 (1 + \lambda V_{DS2})$

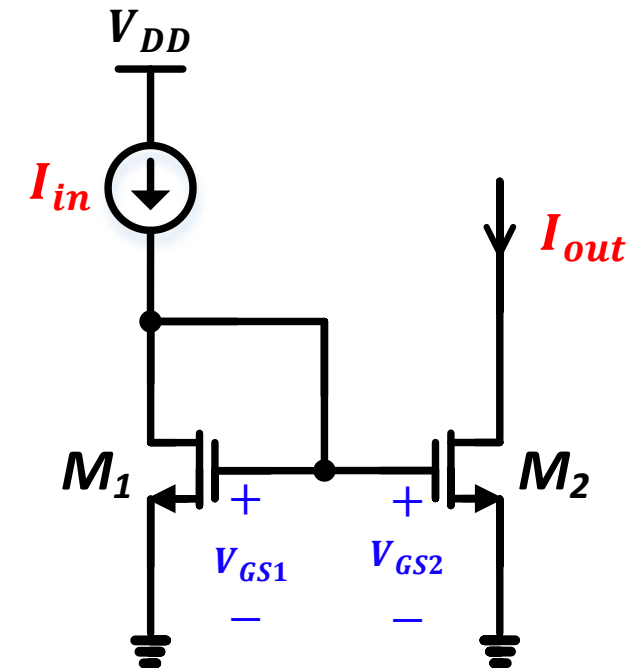
□  $\frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1} \times \left[ \frac{1 + \lambda V_{DS2}}{1 + \lambda V_{DS1}} \right] \rightarrow$  □  $\frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1} \times \left[ 1 + \frac{\lambda (V_{DS2} - V_{DS1})}{1 + \lambda V_{DS1}} \right]$

□ Generally,  $\lambda V_{DS1} \ll 1$

□  $\frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1} \times [1 + \lambda (V_{DS2} - V_{DS1})] = \frac{(W/L)_2}{(W/L)_1} \times [1 + \epsilon]$

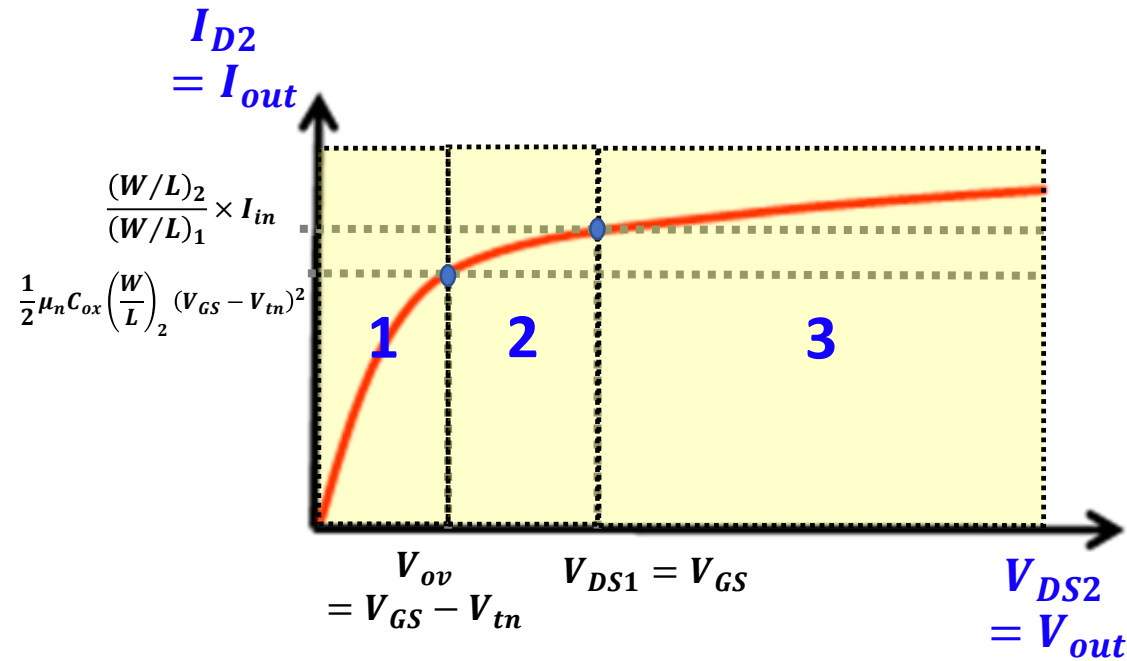
□  $\epsilon$  : is the systematic gain error due to the channel length modulation effect

□  $\epsilon = \lambda (V_{DS2} - V_{DS1})$



# Simple MOSFET Current Mirror

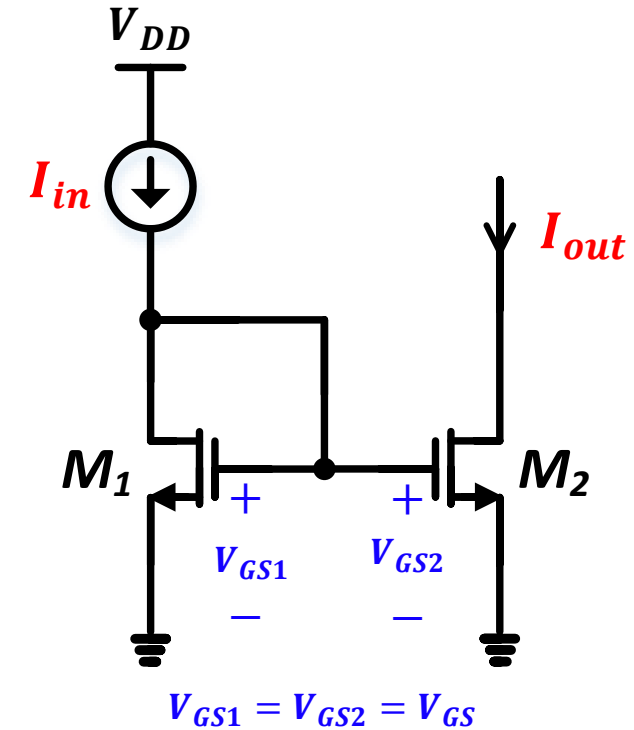
$$\square \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1} \times [1 + \epsilon] \quad \square \epsilon = \lambda (V_{DS2} - V_{DS1})$$



$\square$  @ [1]  $\rightarrow V_{out} < V_{ov} \rightarrow$  **M2 in triode**,  $I_{out} \ll \left[ \frac{(W/L)_2}{(W/L)_1} \times I_{in} \right]$

$\square$  @ [2]  $\rightarrow V_{ov} < V_{out} < V_{DS1} \rightarrow$  **M2 in saturation with -ve  $\epsilon$** ,  $I_{out} < \left[ \frac{(W/L)_2}{(W/L)_1} \times I_{in} \right]$

$\square$  @ [3]  $\rightarrow V_{DS2} > V_{DS1} \rightarrow$  **M2 in saturation with +ve  $\epsilon$** ,  $I_{out} > \left[ \frac{(W/L)_2}{(W/L)_1} \times I_{in} \right]$



$\square$   $M_1 \rightarrow$  G & D are shorted [forced to be in saturation]

$$\square V_{DS1} = V_{GS1}$$

# Simple MOSFET Current Mirror

$$\square \frac{I_{out}}{I_{in}}$$

$$\square \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1} \times [1 + \epsilon]$$

$$\square \epsilon = \lambda (V_{DS2} - V_{DS1})$$



$$\square V_{IN}$$

$$\square V_{IN} = V_{GS} = V_{ov} + V_{tn}$$



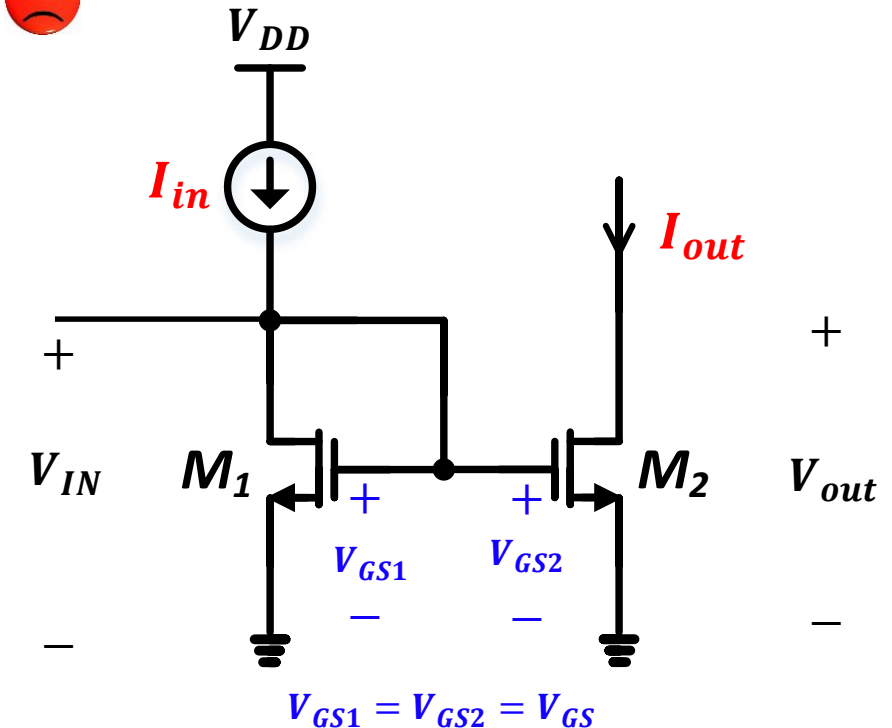
$$\square V_{out,min}$$

$$\square V_{out,min} = V_{ov}$$



$$\square R_{out}$$

$$\square R_{out} = r_{o2}$$



# MOSFET Cascode Current Mirror

□ Get  $\frac{I_{out}}{I_{in}}$

□  $I_G = 0, V_{GS1} = V_{GS2} = V_{GS} \rightarrow V_{ov1} = V_{ov2}, I_{in} = I_1 = I_3 \text{ \& } I_{out} = I_2 = I_4$

- ❑ Assume that (M3 & M1) and (M2 & M4) are matched  $\rightarrow$  All transistors have the same overdrive voltage ( $V_{ov}$ )

$$\square \quad \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1} \times [1 + \epsilon]$$

$$\square \quad \epsilon = \lambda (V_{DS2} - V_{DS1})$$

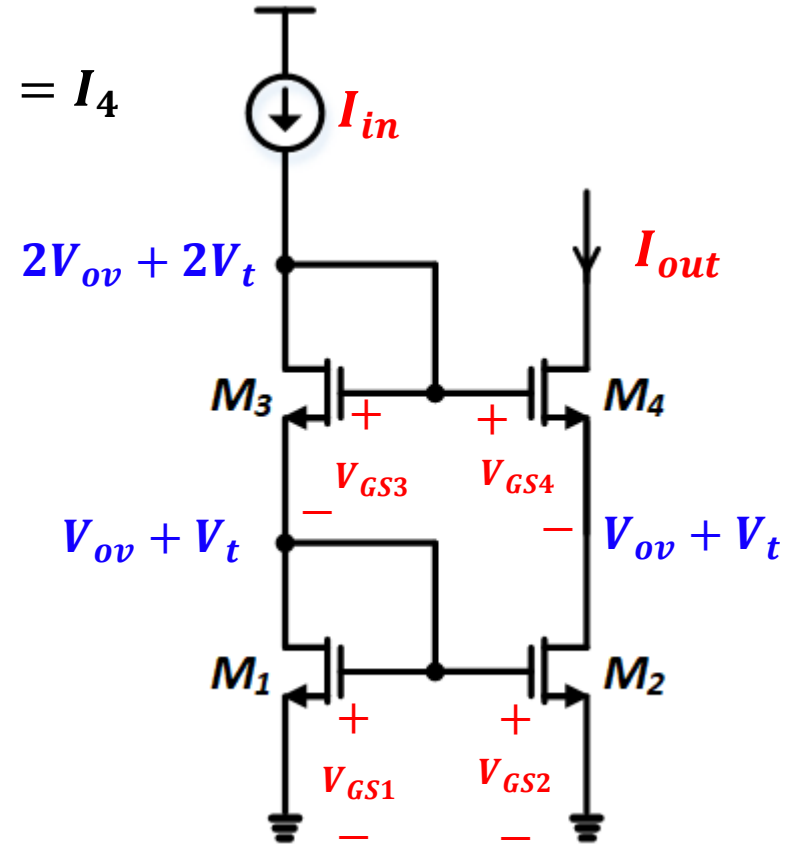
$$\square \quad V_{DS1} = V_{GS1} = V_{ov1} + V_{t1} = V_{ov} + V_t$$

$$\square \quad V_{DS2} = V_{GS1} + V_{GS3} - V_{GS4} = V_{ov} + V_t$$

□ As  $V_{DS1} = V_{DS2}$ ,  $\epsilon \approx 0 \rightarrow$

$$\square \quad \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1}$$

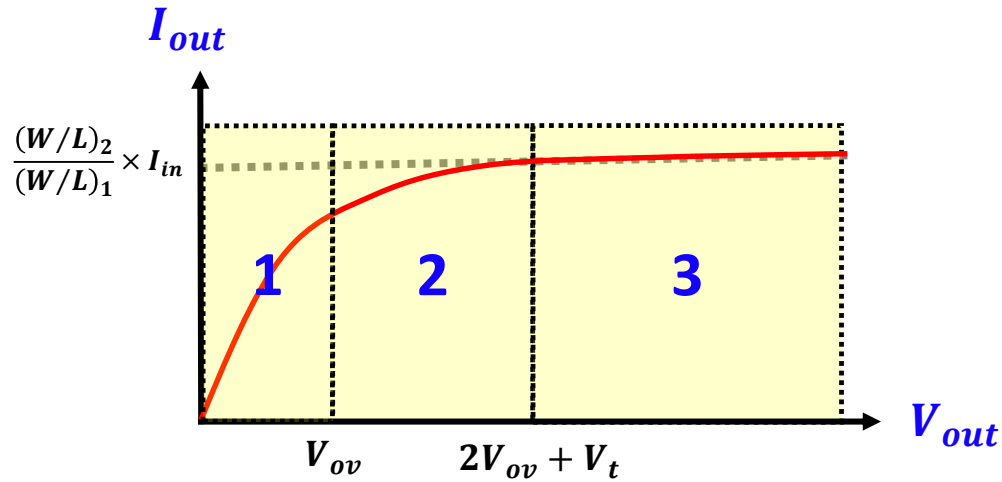
**□  $\epsilon \approx 0$**



# MOSFET Cascode Current Mirror

$$\square \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1}$$

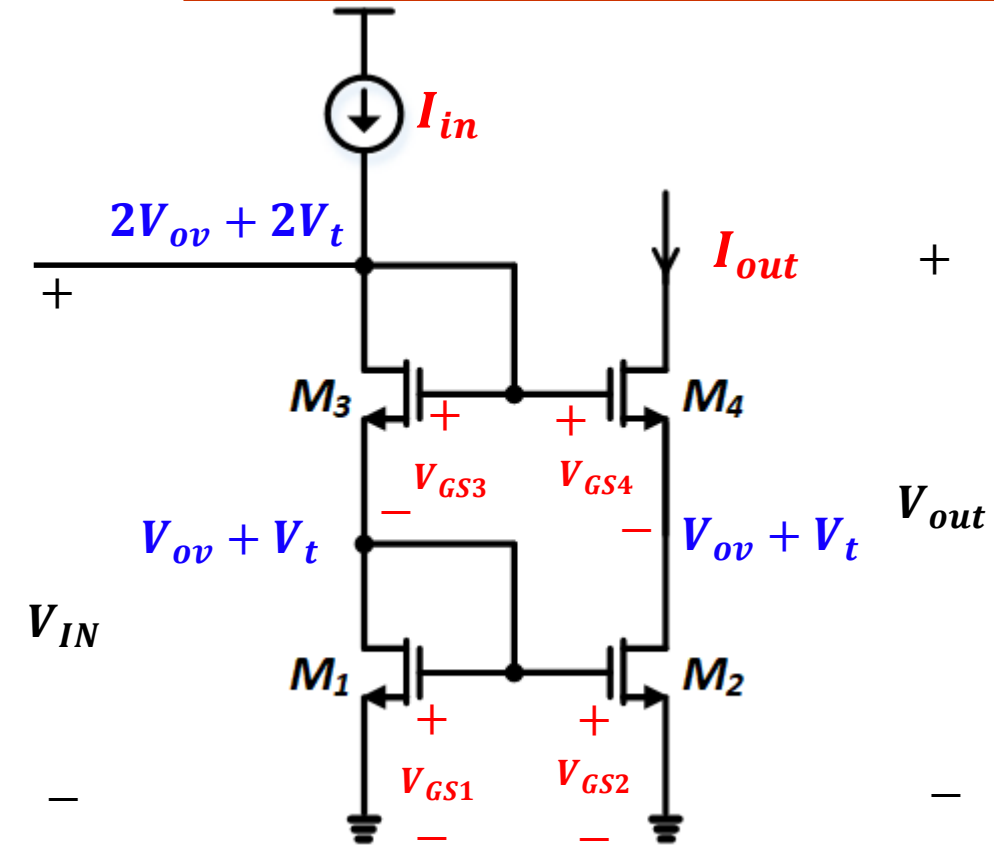
$$\square \epsilon \approx 0$$



$$\square @ [1] \rightarrow V_{out} < V_{ov} \rightarrow \text{M2 \& M4 in triode, } I_{out} \ll \left[ \frac{(W/L)_2}{(W/L)_1} \times I_{in} \right]$$

$$\square @ [2] \rightarrow V_{ov} < V_{out} < 2V_{ov} + V_t \rightarrow \text{M2 in saturation, M4 in triode, } I_{out} < \left[ \frac{(W/L)_2}{(W/L)_1} \times I_{in} \right]$$

$$\square @ [3] \rightarrow V_{out} > 2V_{ov} + V_t \rightarrow \text{M2 \& M4 in saturation, } I_{out} = \left[ \frac{(W/L)_2}{(W/L)_1} \times I_{in} \right]$$



# MOSFET Cascode Current Mirror

$$\square \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1}$$

$$\square \epsilon \approx 0$$



$$\square V_{IN} = 2V_{ov} + 2V_t$$

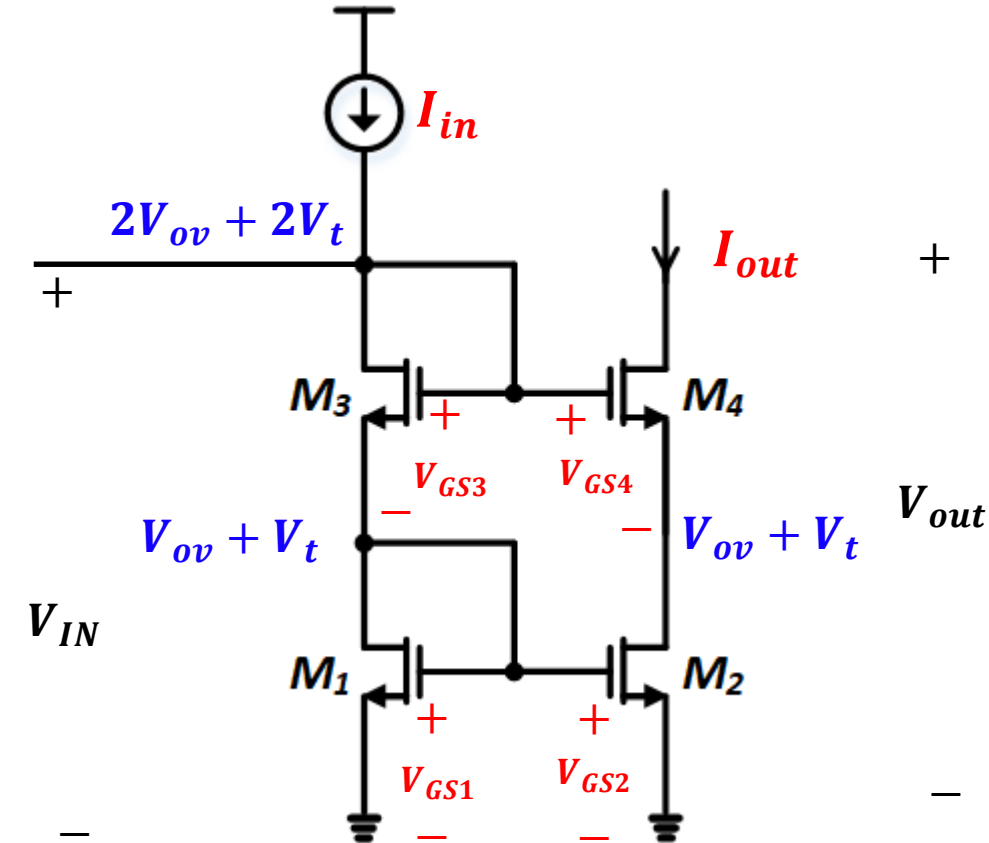


$$\square V_{out,min} = V_{DS2} + V_{DS4,min} = (V_{ov} + V_t) + V_{ov}$$

$$\square V_{out,min} = 2V_{ov} + V_t$$



$$\square R_{out} = r_{o2} + r_{o4} + g_{m4}r_{o2}r_{o4} \approx g_{m4}r_{o2}r_{o4}$$



# Modified Cascode Current Mirror

❑ Here,  $V_{DS2}$  is reduced to  $\approx V_{ov}$ , consequently  $V_{out,min}$  is reduced to  $2V_{ov}$

❑ Assume that (M3 & M1) and (M2 & M4) are matched  $\rightarrow$  All transistors have the same overdrive voltage ( $V_{ov}$ )

❑  $I_G = 0$ ,  $V_{GS1} = V_{GS2} = V_{GS} \rightarrow V_{ov1} = V_{ov2}$ ,  $I_{in} = I_1 = I_3$  &  $I_{out} = I_2 = I_4$

❑ At  $V_B \approx 2V_{ov} + V_t \rightarrow V_{DS1} = V_{DS2} \approx V_{ov} \rightarrow \epsilon \approx 0$

$$\frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1}$$

$$\epsilon \approx 0$$



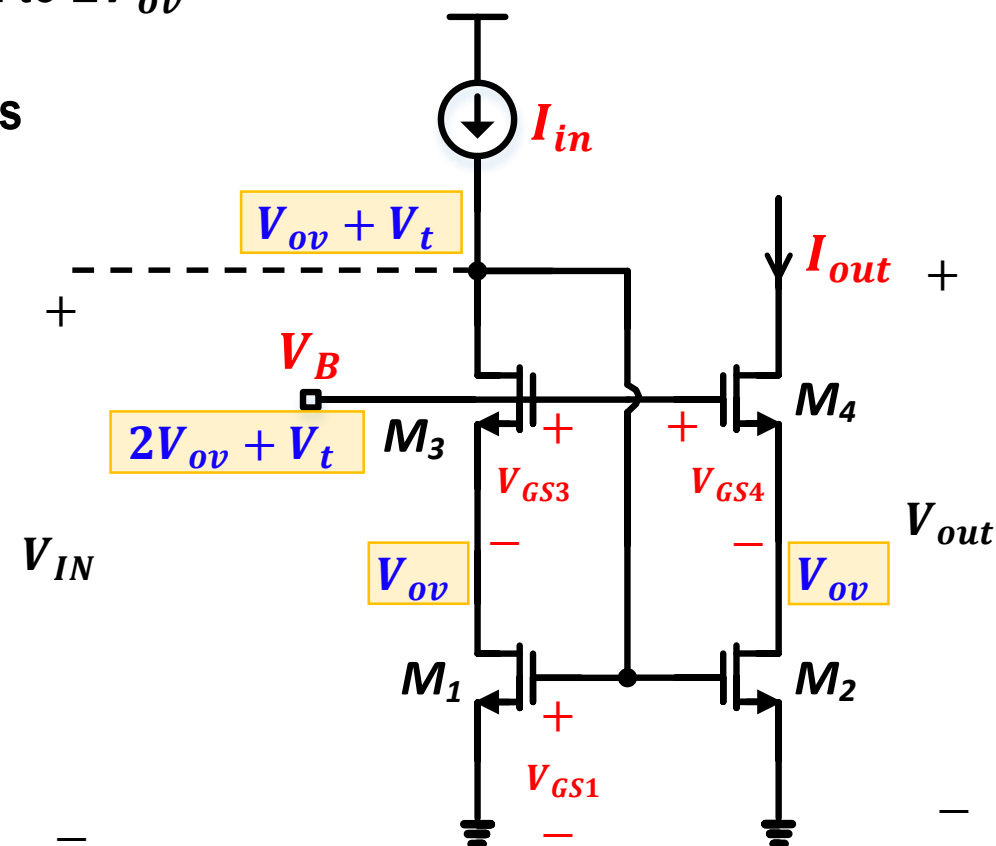
$$V_{IN} = V_{ov} + V_t$$



$$V_{out,min} = 2V_{ov}$$



$$R_{out} = r_{o2} + r_{o4} + g_{m4}r_{o2}r_{o4} \approx g_{m4}r_{o2}r_{o4}$$



# Modified Cascode Current Mirror

❑ Find the range of  $V_B$  such that the modified cascode current mirror work properly (All transistors are in saturation region)

❑ For M1 & M2 to be in saturation

❑  $V_{DS1,2} > V_{ov}$       ❑  $V_B - (V_{ov} + V_t) > V_{ov}$

❑  $V_B > 2V_{ov} + V_t$  (1)

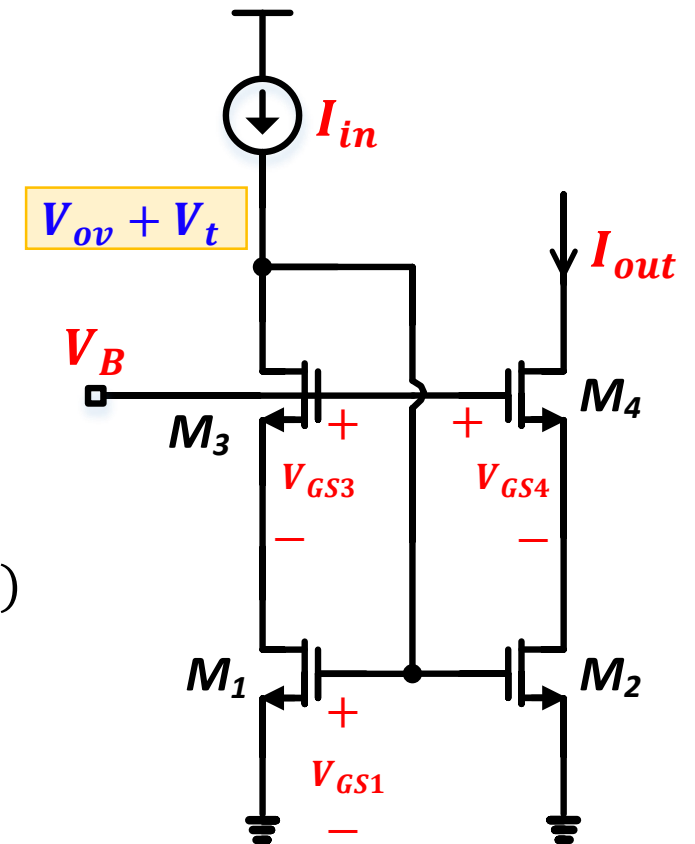
❑ For M3 to be in saturation

❑  $V_{DS3} > V_{ov}$       ❑  $V_{D3} = V_{ov} + V_t$       ❑  $V_{S3} = V_B - (V_{ov} + V_t)$

❑  $V_{DS3} = 2V_{ov} + 2V_t - V_B$

❑  $2V_{ov} + 2V_t - V_B > V_{ov}$       ❑  $V_B < V_{ov} + 2V_t$  (2)

❑  $2V_{ov} + V_t < V_B < V_{ov} + 2V_t$

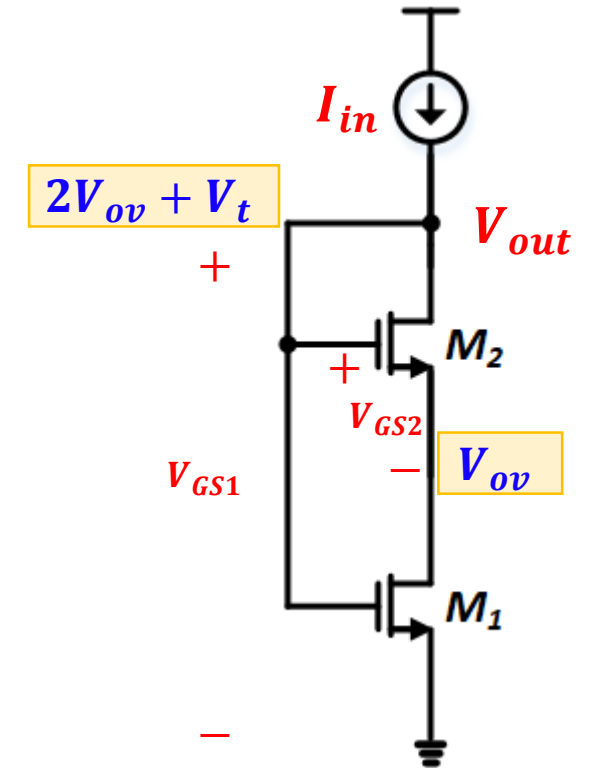


Consequently,  $V_{ov} + 2V_t > 2V_{ov} + V_t$

❑  $V_t > V_{ov}$

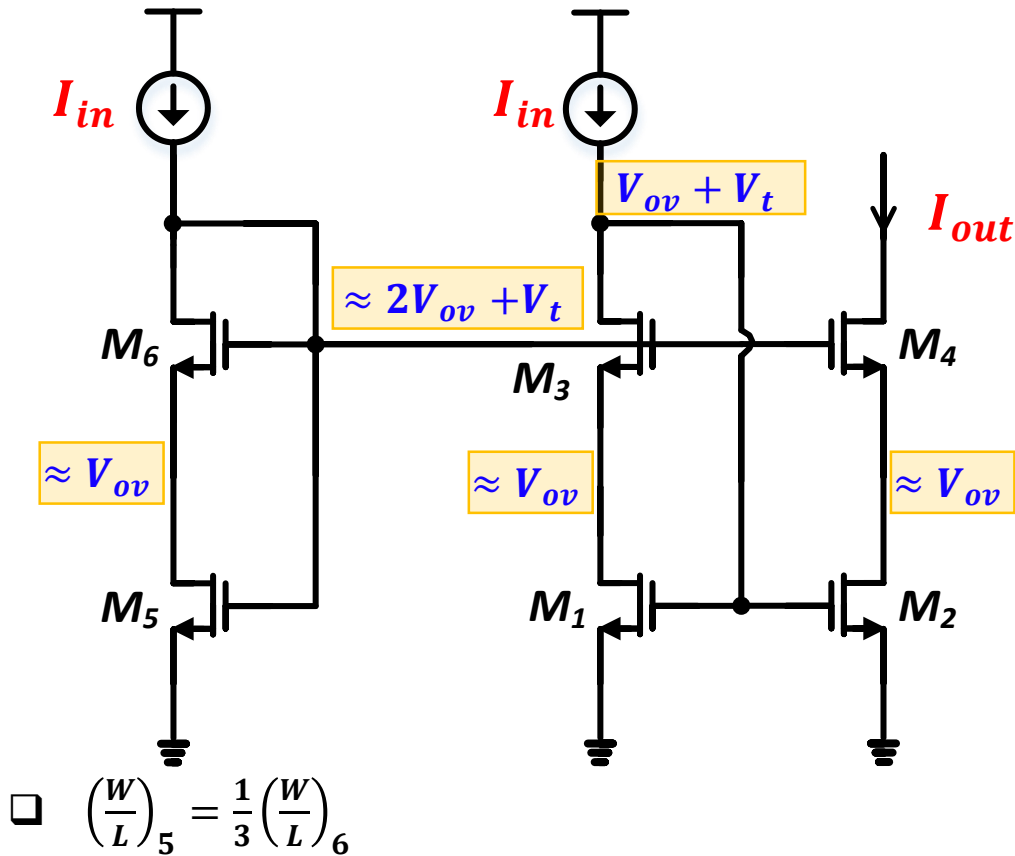
# How to Generate $V_B \approx 2V_{ov} + V_t$

- ❑ The shown circuit could be used to provide  $V_B \approx 2V_{ov} + V_t$
- ❑ Note: **M2** is diode connected and it is forced to operate in **saturation** region with  $V_{GS2} = V_{ov} + V_t$
- ❑ As  $V_{out} = V_{GS1} \approx 2V_{ov} + V_t$ ,  $V_{GS1} = V_{GS2} + V_{DS1} \rightarrow V_{DS1} \approx V_{ov}$
- ❑ Check operating region of M1:
  - ❑  $V_{GS1} \approx 2V_{ov} + V_t$
  - ❑  $V_{DS1} \approx V_{ov}$
  - ❑  $V_{DS1} < V_{GS1} - V_t$
  - ❑ **M1** operates in **triode** region
- ❑ Get the required relationship between  $(W/L)_1$  and  $(W/L)_2$ 
  - ❑  $I_{in} = I_1 = I_2$
  - ❑  $\mu_n C_{ox} \left(\frac{W}{L}\right)_1 \left[ (V_{GS1} - V_t) V_{DS1} - \frac{V_{DS1}^2}{2} \right] = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_2 [V_{GS2} - V_t]^2$
  - ❑  $\left(\frac{W}{L}\right)_1 \left[ 2V_{ov}^2 - \frac{V_{ov}^2}{2} \right] = \frac{1}{2} \left(\frac{W}{L}\right)_2 V_{ov}^2$       ❑  $\left(\frac{W}{L}\right)_1 [1.5 V_{ov}^2] = \frac{1}{2} \left(\frac{W}{L}\right)_2 V_{ov}^2$



$$\boxed{\left(\frac{W}{L}\right)_1 = \frac{1}{3} \left(\frac{W}{L}\right)_2}$$

# High Swing Current Mirror with Two input Branches



- 



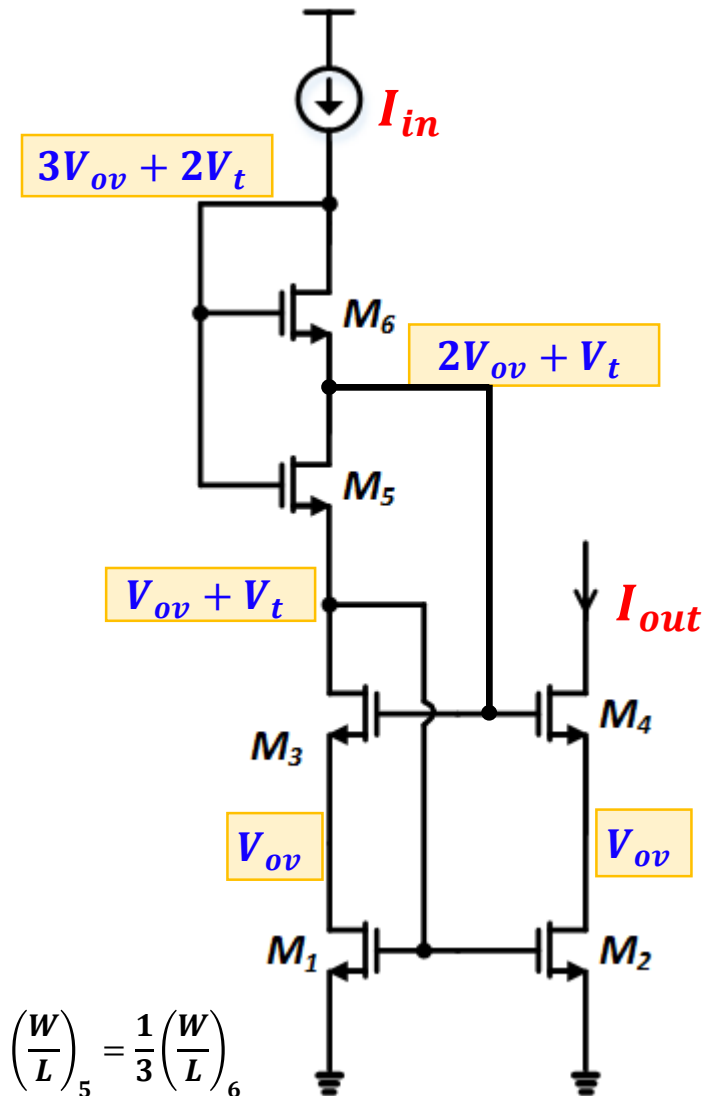




- 

- ❑ Can we use only one input branch??**

# Sooch Cascode Current Mirror



❑ Modified cascode current mirror with only one branch to generate all required biasing.

❑ As  $V_{DS1} \approx V_{DS2} \approx V_{ov}$

$$\square \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1}$$

$$\square \epsilon \approx 0$$



$$\square V_{IN} = 3V_{ov} + 2V_t$$



$$\square V_{out,min} = 2V_{ov}$$



$$\square R_{out} \approx g_{m4} r_{o2} r_{o4}$$



# Wilson Current Mirror (MOSFET)

❑ Feedback is used to stabilize the output current and provide a high output impedance

❑ As  $V_{out} \uparrow \Rightarrow I_{out} \uparrow \Rightarrow V_1 \uparrow \Rightarrow V_2 \downarrow \Rightarrow V_3 \downarrow \Rightarrow I_{out} \downarrow$

❑ As  $V_{DS1} \approx V_{DS2} \approx V_{ov}$

$$\square \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1}$$

$$\square \epsilon \approx 0$$



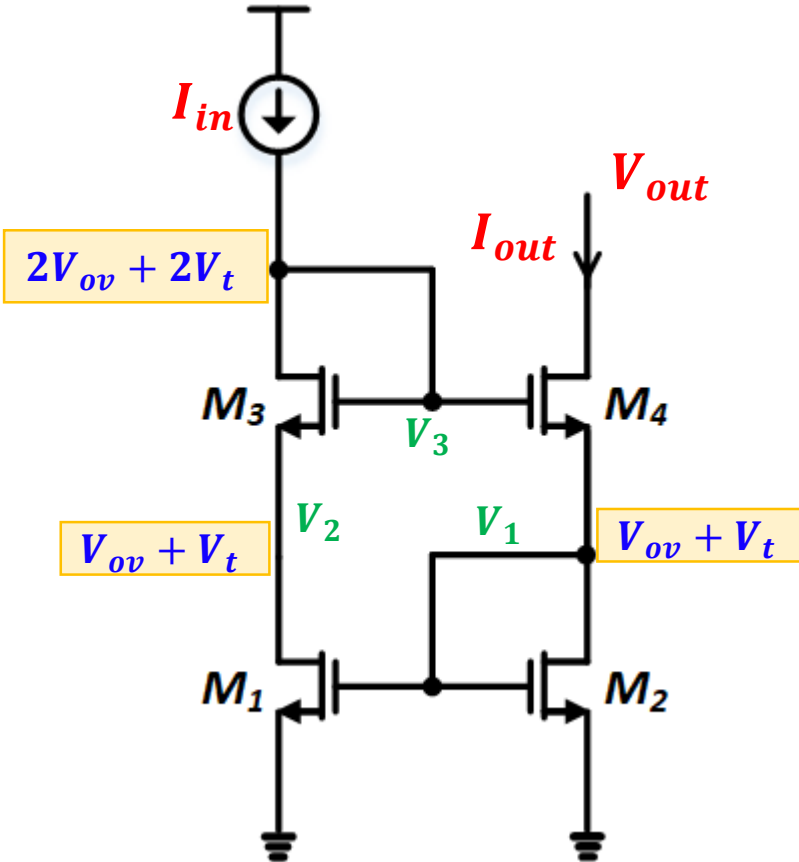
$$\square V_{IN} = 2V_{ov} + 2V_t$$



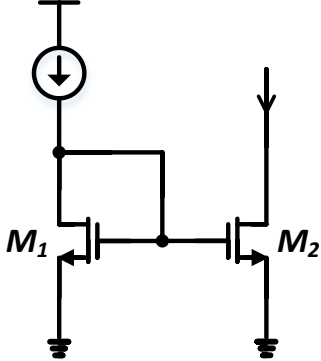
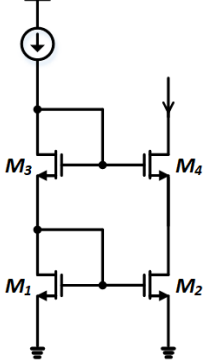
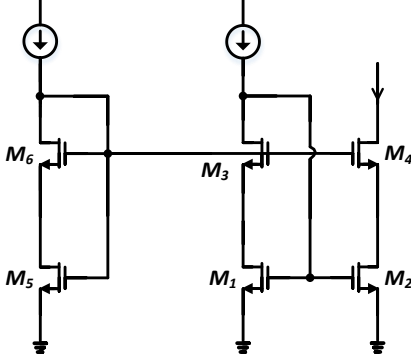
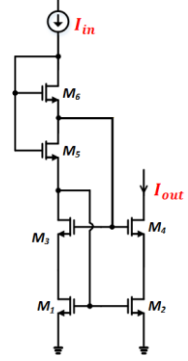
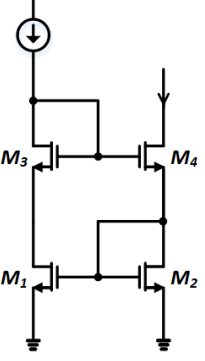
$$\square V_{out,min} = 2V_{ov} + V_t$$



$$\square R_{out} \approx r_{o2} + r_{o4} + g_{m4}r_{o2}r_{o4} \approx g_{m4}r_{o2}r_{o4}$$



# Summary of Basic MOSFET Current Mirrors

|               | Simple                                                                            | Cascode                                                                            | High Swing                                                                          | Sooch                                                                               | Wilson                                                                              |
|---------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|               |  |  |  |  |  |
| $V_{IN}$      | $V_{ov} + V_t$ 😊                                                                  | $2V_{ov} + 2V_t$ 😞                                                                 | $2V_{ov} + V_t$<br>$V_{ov} + V_t$ 😊                                                 | $3V_{ov} + 2V_t$ 😞                                                                  | $2V_{ov} + 2V_t$ 😞                                                                  |
| $V_{out,min}$ | $V_{ov}$ 😊                                                                        | $2V_{ov} + V_t$ 😞                                                                  | $2V_{ov}$ 😊                                                                         | $2V_{ov}$ 😊                                                                         | $2V_{ov} + V_t$ 😞                                                                   |
| $R_{out}$     | $r_o$ 😞                                                                           | $\approx g_m r_o^2$ 😊                                                              | $\approx g_m r_o^2$ 😊                                                               | $\approx g_m r_o^2$ 😊                                                               | $\approx g_m r_o^2$ 😊                                                               |
| $\epsilon$    | $\lambda(V_{DS2} - V_{DS1})$ 😞                                                    | $\approx 0$ 😊                                                                      | $\approx 0$ 😊                                                                       | $\approx 0$ 😊                                                                       | $\approx 0$ 😊                                                                       |

# Outline

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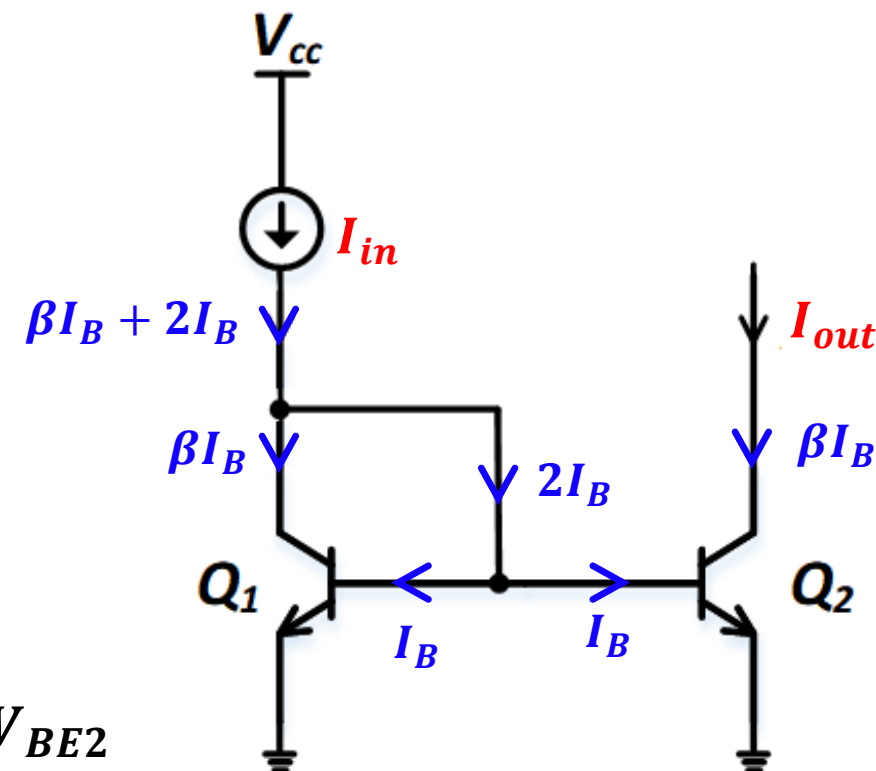
- ❑ Amplifiers (Get the gain using  $G_m \times R_{out}$ )
- ❑ Current Mirrors
  - ❑ Ideal Current Source
  - ❑ Transistor as Current Source
  - ❑ Current Mirror Specifications
  - ❑ MOSFET Current Mirrors
  - ❑ **BJT Current Mirrors (Simple, Buffered, Cascode, Wilson, Wildlar)**

# Simple BJT Current Mirror

- ❑ For BJT transistor:  $I_C = I_S e^{(V_{BE}/V_T)} \times \left[ 1 + \frac{V_{CE}}{V_A} \right]$
- ❑ Ignore the effect of early voltage ( $V_A = \infty, r_o = \infty$ )
- ❑  $I_C = I_S e^{(V_{BE}/V_T)}$
- ❑ Q1 and Q2 are matches  $\rightarrow$  have the same  $\beta$  and the same emitter area ( $A_E$ ) [Consequently have the same  $I_S$ ].

❑ Get  $\frac{I_{out}}{I_{in}}$

- ❑ As Q1 & Q2 are matched  $\rightarrow$  same  $I_S$  and  $\beta$  and  $V_{BE1} = V_{BE2}$
- ❑  $I_{B1} = I_{B2} = I_B$       ❑  $I_{C1} = I_{C2} = \beta I_B$
- ❑ From The circuit, get  $I_{in}$  and  $I_{out}$



# Simple BJT Current Mirror

□ Get  $\frac{I_{out}}{I_{in}}$

□  $I_{in} = \beta I_B + 2 I_B = I_B(\beta + 2)$

□  $I_{out} = \beta I_B$

□  $\frac{I_{out}}{I_{in}} = \frac{\beta}{\beta+2} = 1 - \left[ \frac{2}{\beta+2} \right] = 1 - \epsilon$

□  $\epsilon$  : is the systematic gain error due to the finite value of  $\beta$

□  $V_{IN}$

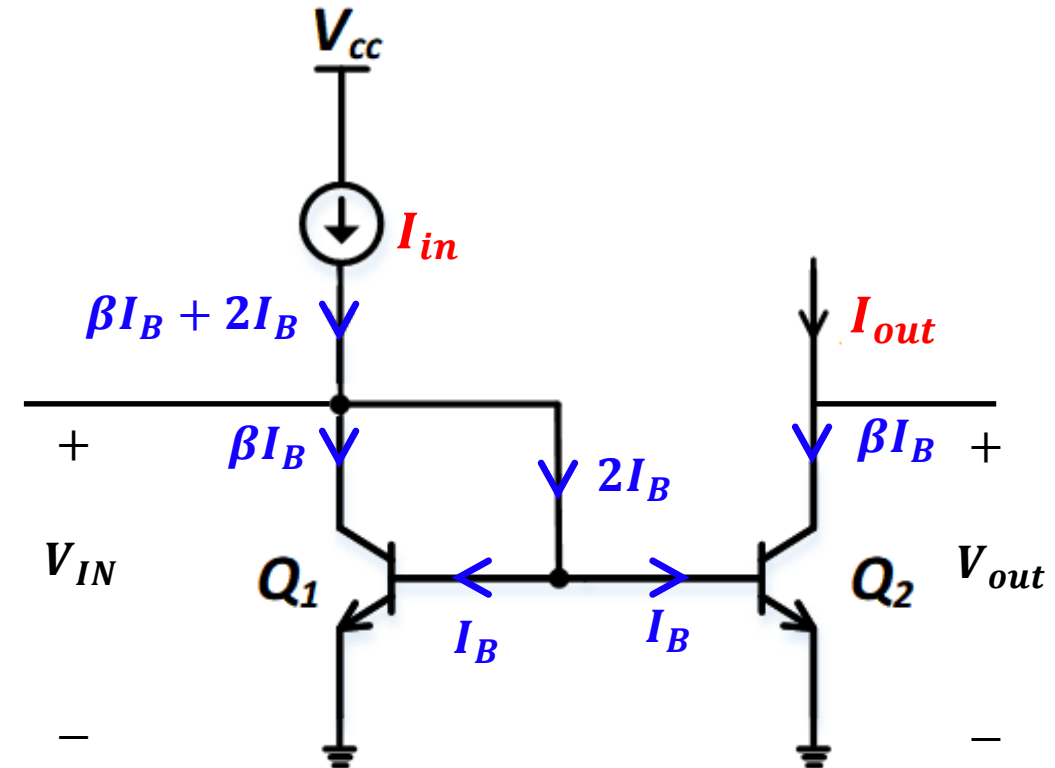
□  $V_{IN} = V_{BE} \approx 0.7 V$  😊

□  $V_{out,min}$

□  $V_{out,min} = V_{CE2,sat} \approx 0.2 V$  😊

□  $R_{out}$

□  $R_{out} = r_{o2}$  😞



□  $\epsilon$

□  $\epsilon = \frac{2}{\beta+2}$  😞

# Simple BJT Current Mirror

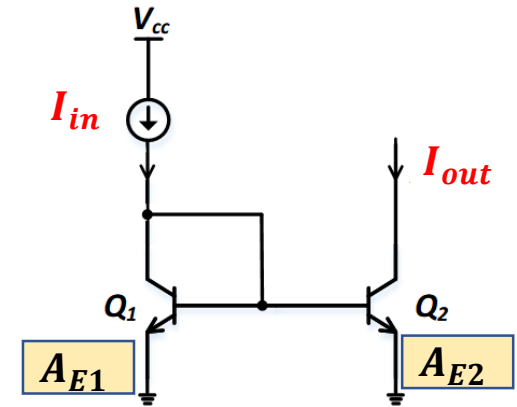
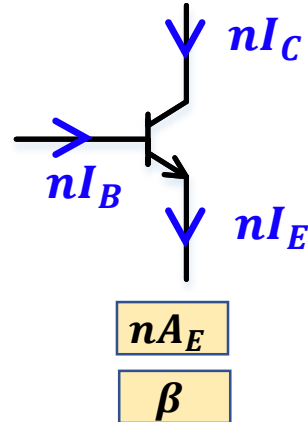
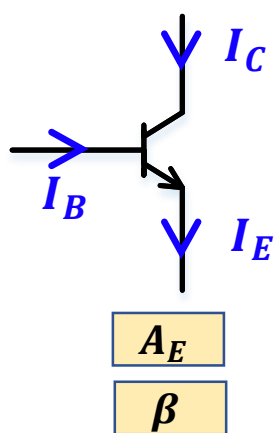
□ How to get adjust the mirror ratio

□  $I_C = I_S e^{(V_{BE}/V_T)}$

□  $I_S \propto A_E$  (Emitter area)

□ Mirror ratio =  $\frac{I_{out}}{I_{in}} \propto \frac{A_{E2}}{A_{E1}}$

□ Note:



□ Similar to connecting  $n$  parallel devices

□ Beta remains constant.

# Simple BJT Current Mirror

❑ Get  $\frac{I_{out}}{I_{in}}$

$$\square \quad I_{in} = \beta I_B + I_B(n+1) = I_B(\beta + n+1)$$

$$\square \quad I_{out} = n\beta I_B$$

$$\square \quad \frac{I_{out}}{I_{in}} = \frac{n\beta}{\beta+n+1} = n \left[ 1 - \frac{n+1}{\beta+n+1} \right] = n[1 - \epsilon]$$

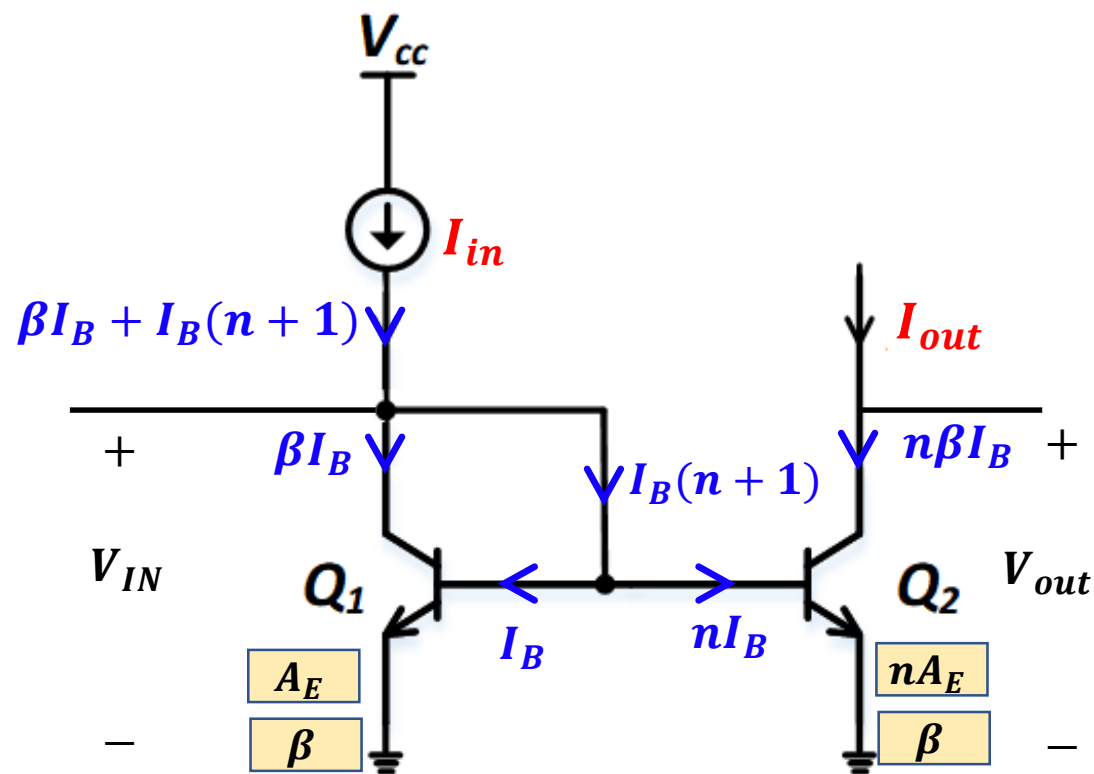
□ **Note: as  $n$  increases,  $\epsilon$  increases.**

□  $V_{IN} = V_{BE1} \approx 0.7 \text{ V}$

□  $V_{out,min} = V_{CE2,sat} \approx 0.2 V$

□  $R_{out} = r_{o2}$

$$\square \quad \epsilon = \frac{n+1}{\beta+n+1}$$



# Buffered BJT Current Mirror

❑ Transistor Q3 acts as a buffer and reduces the drawn current from the input branch, leading to a smaller  $\epsilon$

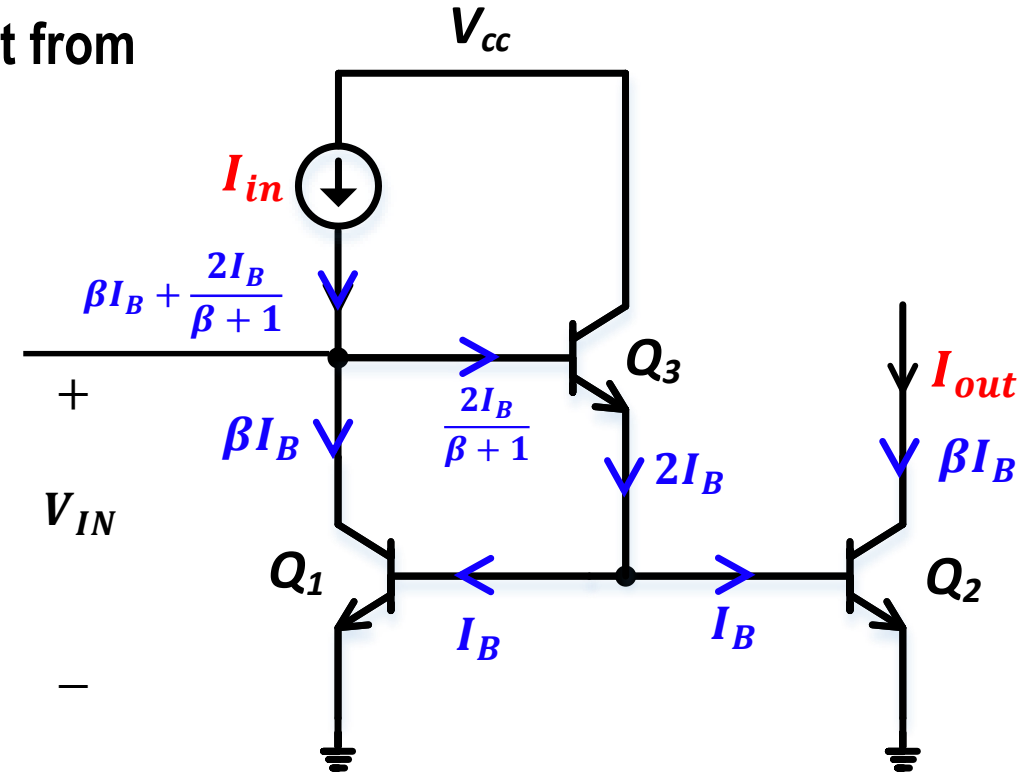
❑ Q1, Q2, and Q3 have the same beta and emitter area.

❑ Get  $\frac{I_{out}}{I_{in}}$

$$\square I_{in} = \beta I_B + \frac{2I_B}{\beta+1} = I_B \frac{\beta(\beta+1)+2}{\beta+1} = I_B \frac{\beta^2 + \beta + 2}{\beta+1}$$

$$\square I_{out} = \beta I_B$$

$$\square \frac{I_{out}}{I_{in}} = \frac{\beta(\beta+1)}{\beta^2 + \beta + 2} = \frac{\beta^2 + \beta}{\beta^2 + \beta + 2} = 1 - \left[ \frac{2}{\beta^2 + \beta + 2} \right] = 1 - \epsilon$$



$$\square V_{IN} = V_{BE1} + V_{BE3} \approx 1.4 V \quad \text{😬}$$

$$\square V_{out,min} = V_{CE2,sat} \approx 0.2 V \quad \text{😄}$$

$$\square R_{out} = r_{o2} \quad \text{😞}$$

$$\square \epsilon = \frac{2}{\beta^2 + \beta + 2} \quad \text{😄}$$

# Cascode Current Mirror (BJT)

Q1, Q2, Q3, and Q4 have the same beta and emitter area.

$\frac{I_{out}}{I_{in}}$

$$I_{in} = I_B \frac{\beta^2 + 2\beta}{\beta + 1} + I_B \frac{2\beta + 2}{\beta + 1} = I_B \frac{\beta^2 + 4\beta + 2}{\beta + 1}$$

$$I_{out} = I_B \frac{\beta^2}{\beta + 1}$$

$$\frac{I_{out}}{I_{in}} = \frac{\beta^2}{\beta^2 + 4\beta + 2} = 1 - \left[ \frac{4\beta + 2}{\beta^2 + 4\beta + 2} \right] = 1 - \epsilon$$

$$V_{IN} = V_{BE1} + V_{BE3} \approx 1.4 \text{ V}$$



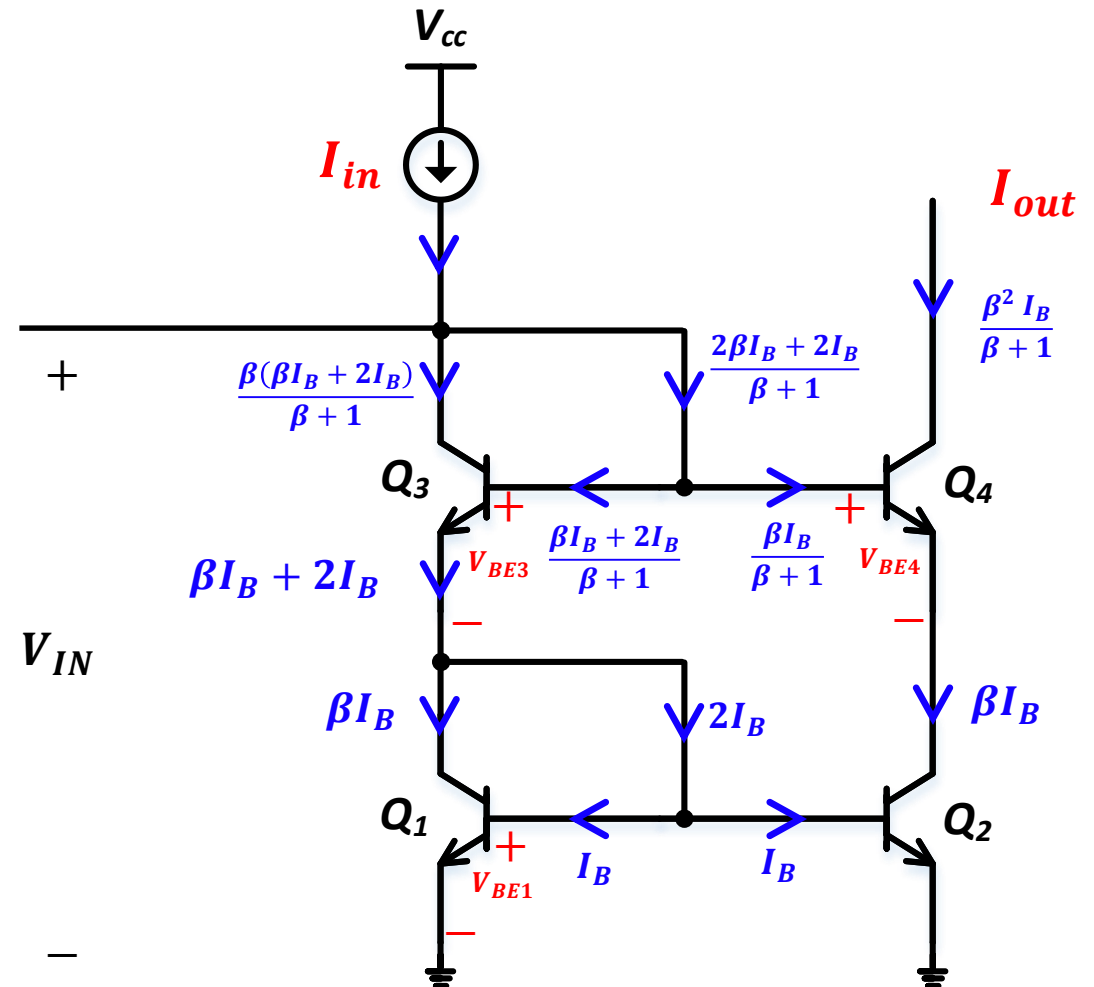
$$V_{out,min} = V_{CE2} + V_{CE4,sat}$$

$$V_{CE2} = V_{BE1} + V_{BE3} - V_{BE4} \approx 0.7 \text{ V}$$

$$V_{out,min} = V_{CE2} + V_{CE4,sat} \approx 0.7 \text{ V} + 0.2 \text{ V} = 0.9 \text{ V}$$



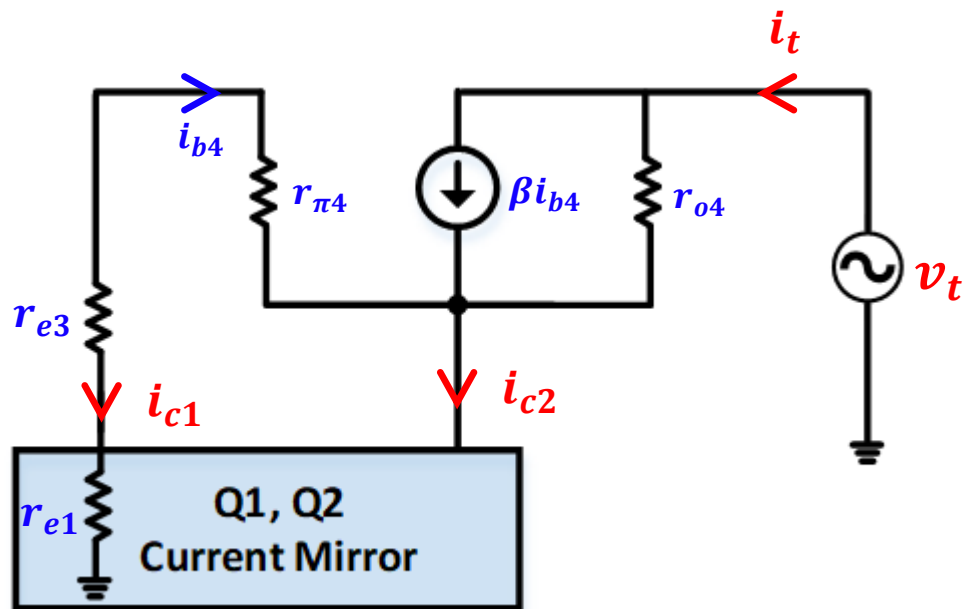
$$\epsilon = \frac{4\beta + 2}{\beta^2 + 4\beta + 2}$$



# Cascode Current Mirror (BJT)

□  $R_{out} = ??$

□ Note: Impedance shortcuts can't be utilized as the base terminals of Q2 and Q4 aren't AC ground. You have to use small signal model



□ From Q1 & Q2 Current Mirror

$$\square i_{c1} = i_{c2}$$

$$\square i_t = i_{c2} + i_{c1} = 2i_{c1} = -2i_{b4}$$

$$\square i_{b4} = \frac{-i_t}{2} \quad \text{①}$$

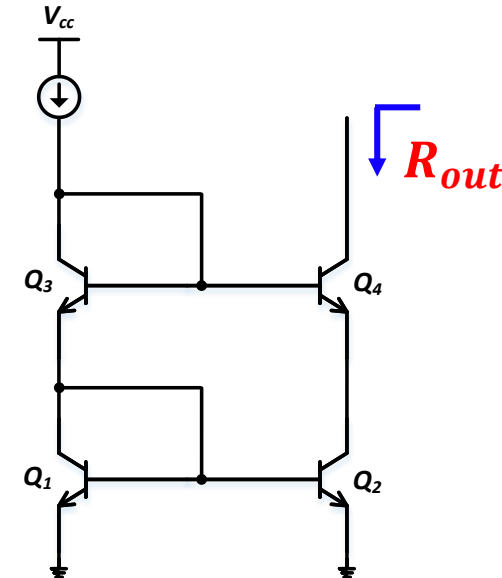
□ Assume  $r_{o4} \gg (r_{\pi4} + r_{e3} + r_{e1})$

□  $\therefore$  approximately, all  $v_t$  appears across  $r_{o4}$

$$\square i_t = \beta i_{b4} + \frac{v_t}{r_{o4}} = \beta \frac{-i_t}{2} + \frac{v_t}{r_{o4}}$$

$$\Rightarrow \square i_t \left(1 + \frac{\beta}{2}\right) = \frac{v_t}{r_{o4}}$$

$$\Rightarrow \square R_{out} = \frac{v_t}{i_t} = \left(1 + \frac{\beta}{2}\right) r_{o4} \approx \frac{\beta r_{o4}}{2}$$



# Wilson Current Mirror (BJT)

- ☐ Q1, Q2, Q3, and Q3 have the same beta and emitter area.

$$\square \quad \frac{I_{out}}{I_{in}}$$

$$\square \quad I_{in} = I_B \frac{\beta^2}{\beta+1} + I_B \frac{2\beta+2}{\beta+1} = I_B \frac{\beta^2 + 2\beta + 2}{\beta+1}$$

$$\blacksquare \quad I_{out} = I_B \frac{\beta^2 + 2\beta}{\beta + 1}$$

$$\square \quad \frac{I_{out}}{I_{in}} = \frac{\beta^2 + 2\beta}{\beta^2 + 2\beta + 2} = 1 - \left[ \frac{2}{\beta^2 + 2\beta + 2} \right] = 1 - \epsilon$$

□  $V_{IN} = V_{BE1} + V_{BE4} \approx 1.4 \text{ V}$



□  $V_{out,min} = V_{BE1} + V_{CE4,sat} \approx 0.7\text{ V} + 0.2\text{ V} \approx 0.9\text{ V}$

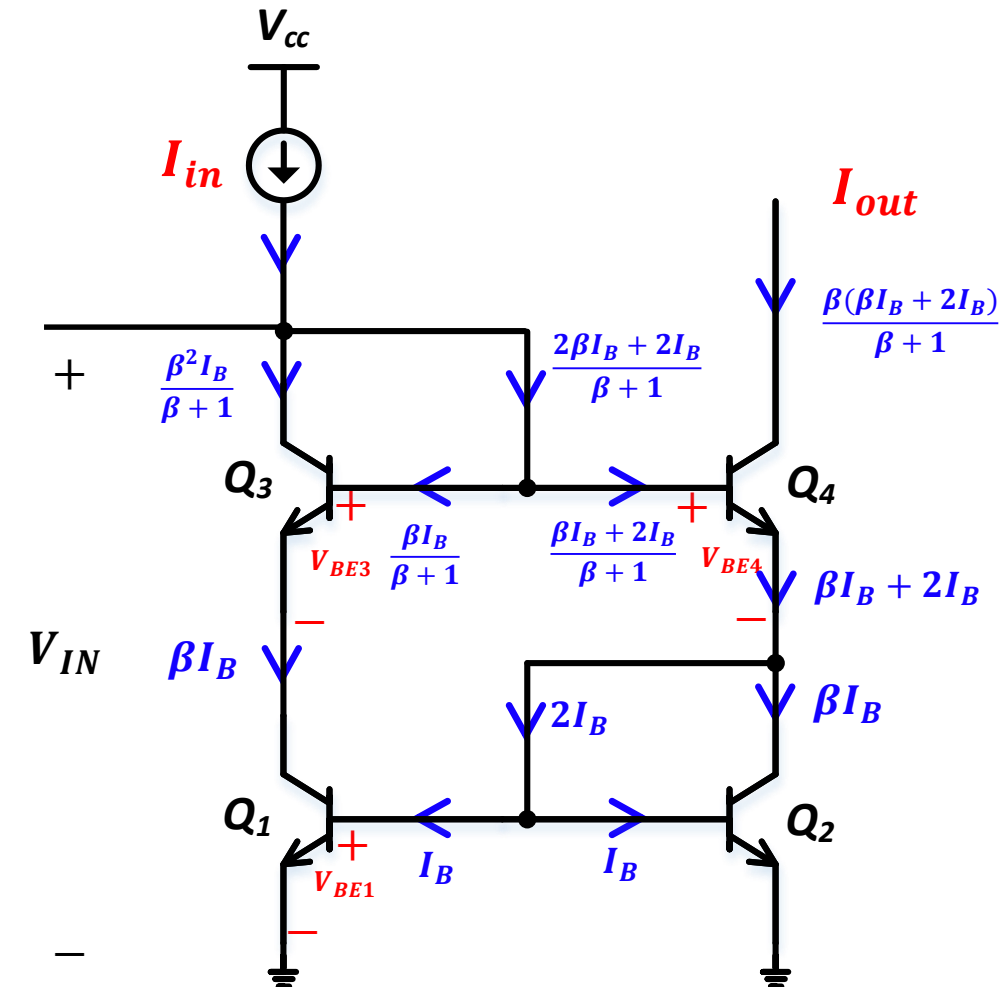


$$\square \quad R_{out} \approx \frac{\beta r_{o4}}{2}$$



## Try to prove it

$$\square \quad \epsilon = \frac{2}{\beta^2 + 2\beta + 2}$$



# Widlar Current Mirror (BJT)

□ Q1 and Q2 are matched (same  $I_S$  and  $\beta$ ). Assume  $\beta = \infty$

□  $\frac{I_{out}}{I_{in}}$

□ As  $\beta = \infty$ ,  $I_{B1} = I_{B2} = 0$

□  $I_C = I_S e^{(V_{BE}/V_T)} \rightarrow V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\right)$

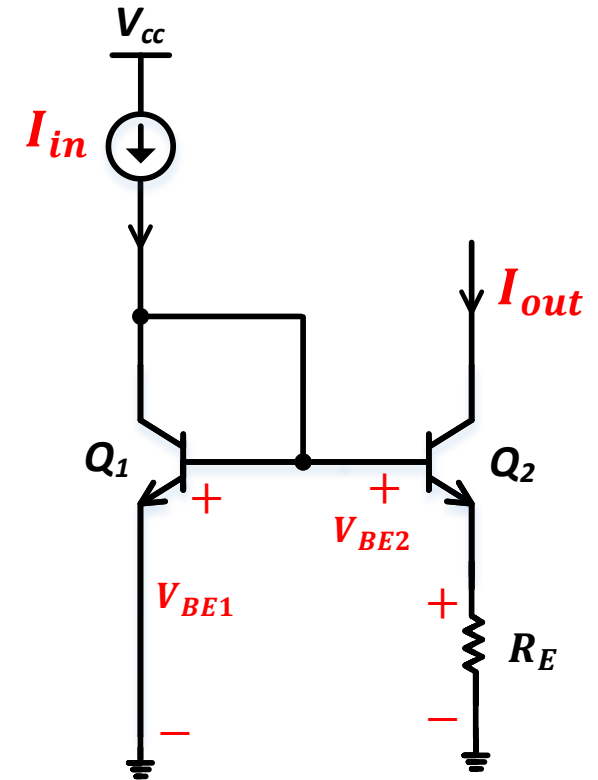
□  $V_{BE1} = V_T \ln\left(\frac{I_{in}}{I_{S1}}\right)$       □  $V_{BE2} = V_T \ln\left(\frac{I_{out}}{I_{S1}}\right)$

□  $V_{BE1} = V_{BE2} + I_{out} R_E \rightarrow$  □  $V_T \ln\left(\frac{I_{in}}{I_{S1}}\right) = V_T \ln\left(\frac{I_{out}}{I_{S2}}\right) + I_{out} R_E$

□  $V_T \ln\left(\frac{I_{in}}{I_{out}} \times \frac{I_{S2}}{I_{S1}}\right) = I_{out} R_E$       At  $I_{S1} = I_{S2}$

□  $V_T \ln\left(\frac{I_{in}}{I_{out}}\right) = I_{out} R_E$

□  $I_{out}$  is controlled by  $I_{in}$  and  $R_E$

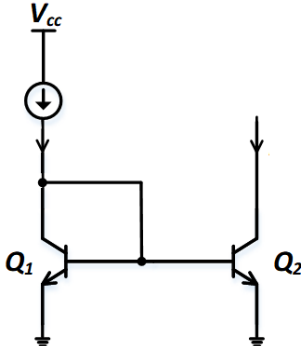
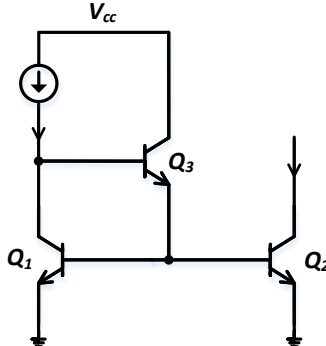
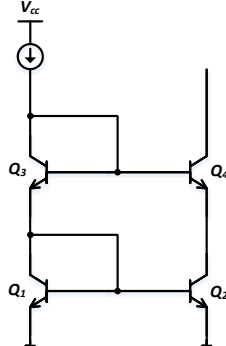
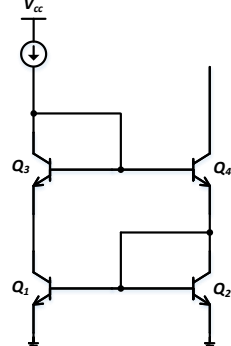
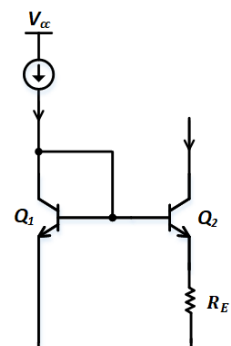


□  $V_{in} = V_{BE1} \approx 0.7 \text{ V}$

□  $V_{out,min} = I_{out} R_E + V_{CE2,sat} \approx I_{out} R_E + 0.2 \text{ V}$

□  $R_{out} \approx r_{o2} [1 + g_m (R_E \parallel r_{\pi})]$

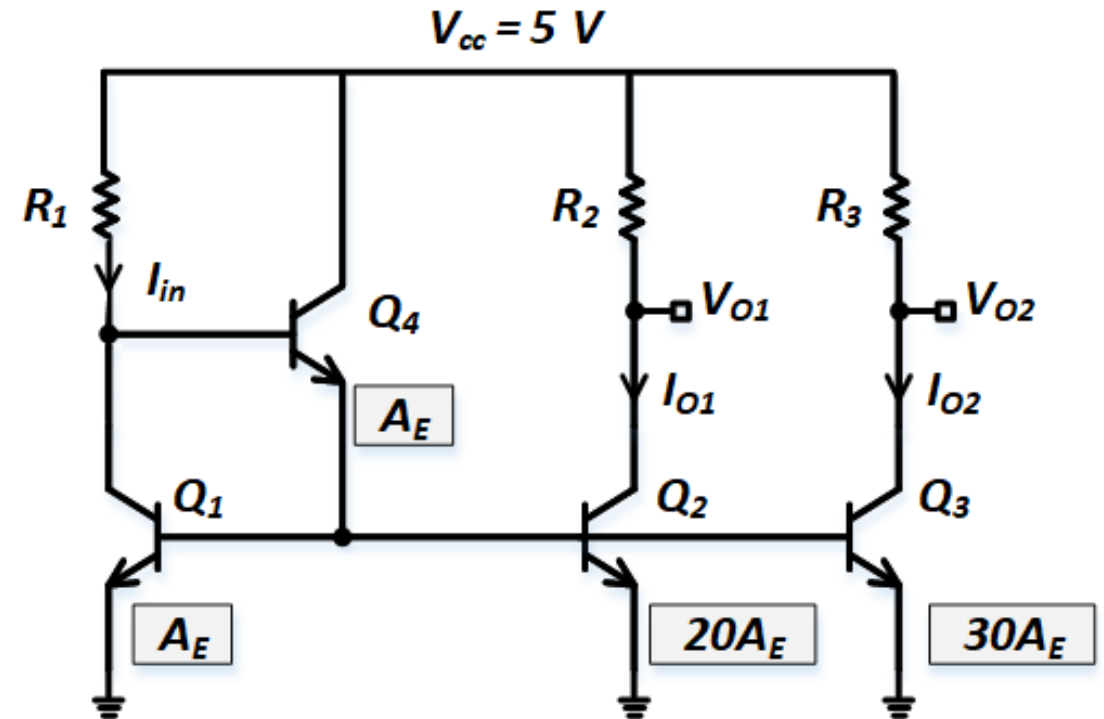
# Summary of Basic BJT Current Mirrors

|               | Simple                                                                            | Buffered                                                                           | Cascode                                                                             | Wilson                                                                              | Widlar                                                                              |
|---------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|               |  |  |  |  |  |
| $V_{IN}$      | $\approx 0.7 V$ 😊                                                                 | $\approx 1.4 V$ 😬                                                                  | $\approx 1.4 V$ 😬                                                                   | $\approx 1.4 V$ 😬                                                                   | $\approx 0.7 V$ 😊                                                                   |
| $V_{out,min}$ | $\approx 0.2 V$ 😊                                                                 | $\approx 0.2 V$ 😊                                                                  | $\approx 0.9 V$ 😬                                                                   | $\approx 0.9 V$ 😬                                                                   | $I_{out}R_E + 0.2 V$ 😊                                                              |
| $R_{out}$     | $r_o$ 😞                                                                           | $r_o$ 😞                                                                            | $\approx \frac{\beta r_o}{2}$ 😊                                                     | $\approx \frac{\beta r_o}{2}$ 😊                                                     | $\approx r_{o2} [1 + g_m R_E]$ 😬                                                    |
| $\epsilon$    | $\frac{2}{\beta + 2}$ 😞                                                           | $\frac{2}{\beta^2 + \beta + 2}$ 😊                                                  | $\frac{4\beta + 2}{\beta^2 + 4\beta + 2}$ 😞                                         | $\frac{2}{\beta^2 + 4\beta + 2}$ 😊                                                  | 😞                                                                                   |

# Example #1 (1/3)

□ For the shown current mirror circuit, all the BJT transistor have the same Beta ( $\beta$ )

- a) Find  $\frac{I_{o2}}{I_{in}}$  and  $\frac{I_{o1}}{I_{in}}$ , and find the corresponding systematic gain error.
- b) If  $\beta = 120$ ,  $R_1 = 3.6 \text{ K}\Omega$ ,  $R_2 = R_3 = 200 \Omega$ , find  $V_{o1}$  and  $V_{o2}$ .



# Example #1 (1/3)

➤ Find  $\frac{I_{o2}}{I_{in}}$  and  $\frac{I_{o1}}{I_{in}}$ , and find the corresponding systematic gain error.

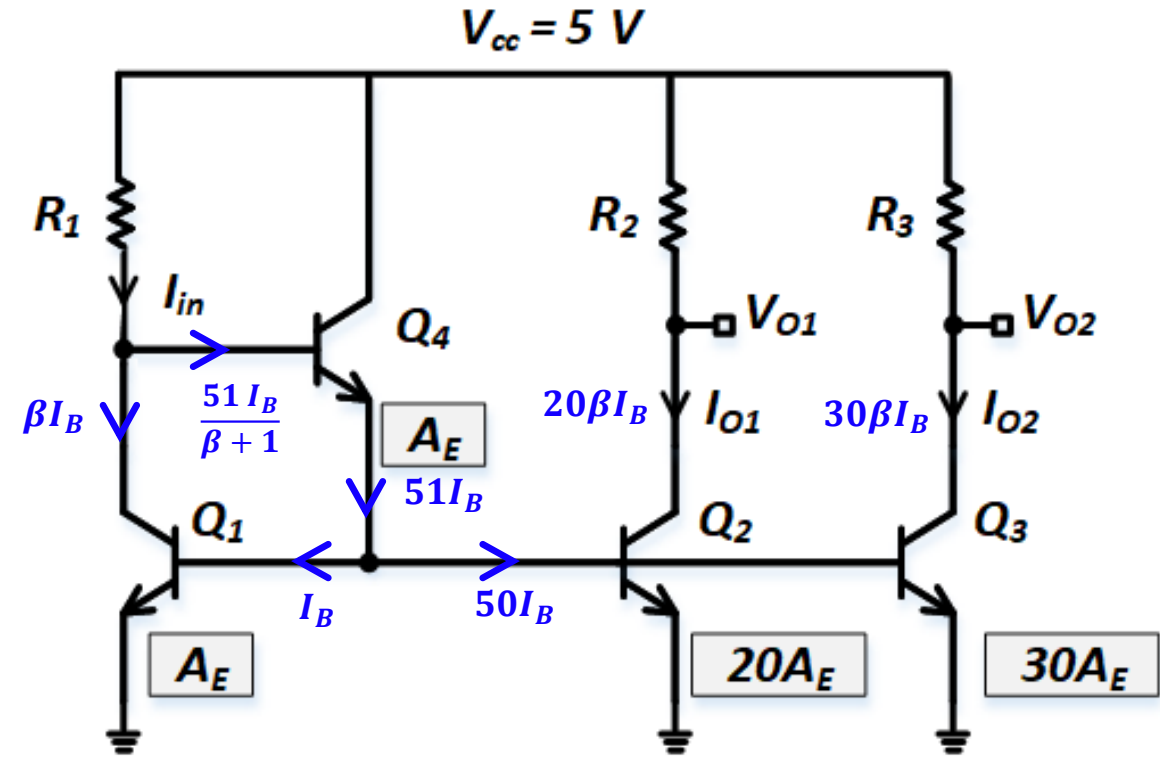
$$\square I_{in} = \beta I_B + \frac{51 I_B}{\beta + 1} = I_B \frac{\beta^2 + \beta + 51}{\beta + 1}$$

$$\square I_{o1} = 20 \beta I_B \quad \square I_{o2} = 30 \beta I_B$$

$$\square \frac{I_{o1}}{I_{in}} = \frac{20 \beta (\beta + 1)}{\beta^2 + \beta + 51} = 20 \frac{\beta^2 + \beta}{\beta^2 + \beta + 51} = 20 \left[ 1 - \frac{51}{\beta^2 + \beta + 51} \right]$$

$$\square \frac{I_{o1}}{I_{in}} = 20[1 - \epsilon] \quad \square \epsilon = \frac{51}{\beta^2 + \beta + 51}$$

$$\square \frac{I_{o2}}{I_{in}} = 30[1 - \epsilon] \quad \square \epsilon = \frac{51}{\beta^2 + \beta + 51}$$



# Example #1 (1/3)

➤ If  $\beta = 120$ ,  $R_1 = 3.6\text{ K}\Omega$ ,  $R_2 = R_3 = 200\text{ }\Omega$ , find  $V_{o1}$  and  $V_{o2}$ .

$$\blacksquare \quad I_{in} = \frac{V_{CC} - V_{BE1} - V_{BE4}}{R_1} = \frac{5 - 0.7 - 0.7}{3.6 \text{ K}} = 1 \text{ mA}$$

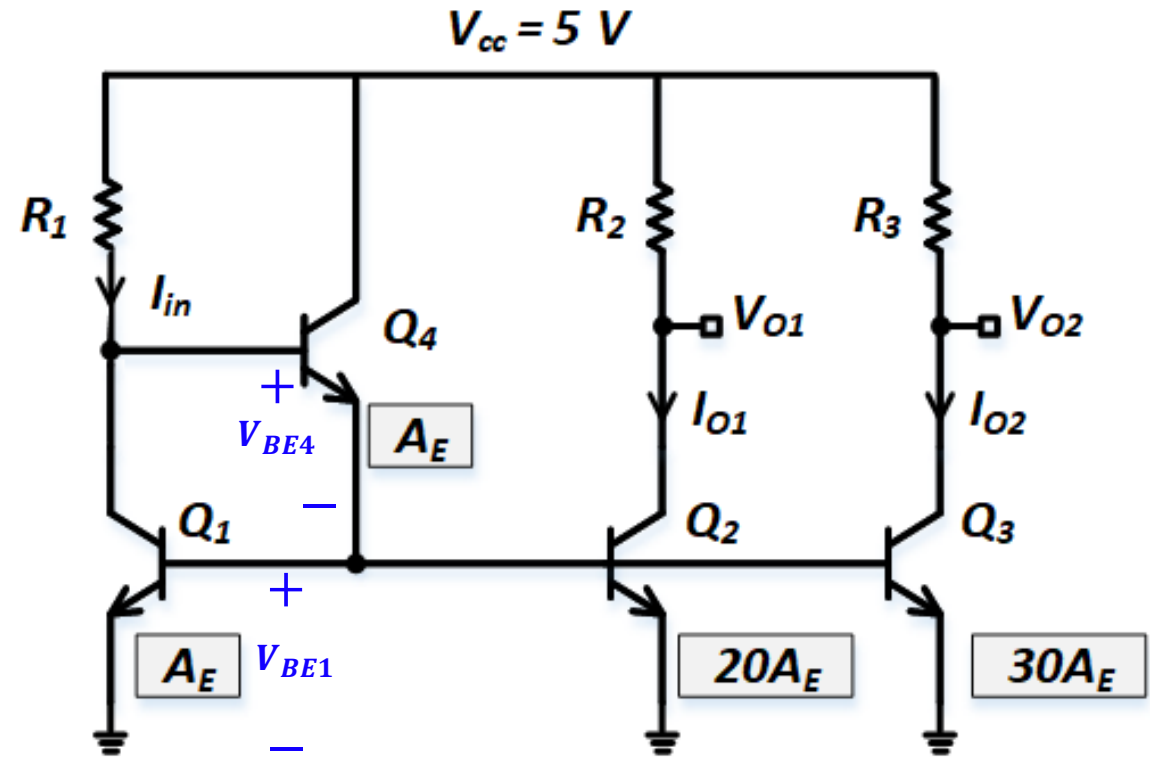
$$\square \quad I_{o1} = I_{in} \times 20 \left[ 1 - \frac{51}{\beta^2 + \beta + 51} \right] = 19.93 \text{ mA}$$

$$\square \quad V_{o1} = V_{CC} - I_{o1}R_2 = 5 - (19.93 \text{ mA})(200\Omega) = 1.014 \text{ V}$$

❑ Check Q2 to be in active:  $V_{CE2} = V_{o1} = 1.014 \text{ V}$ .  
( $V_{CE2} > 0.2 \text{ V}$ ) OK

□  $I_{o2} = I_{in} \times 30 \left[ 1 - \frac{51}{\beta^2 + \beta + 51} \right] = 29.895 \text{ mA}$  Expected output current if Q3 is in active region.

$$\blacksquare \quad V_{o2} = V_{cc} - I_{o2}R_3 = 5 - (29.895 \text{ mA})(200\Omega) = -0.979 \text{ V}$$



❑ As  $V_{CE3} < 0.2 V$ . Q3 is not in active region.  
Solve again considering Q3 in saturation region.

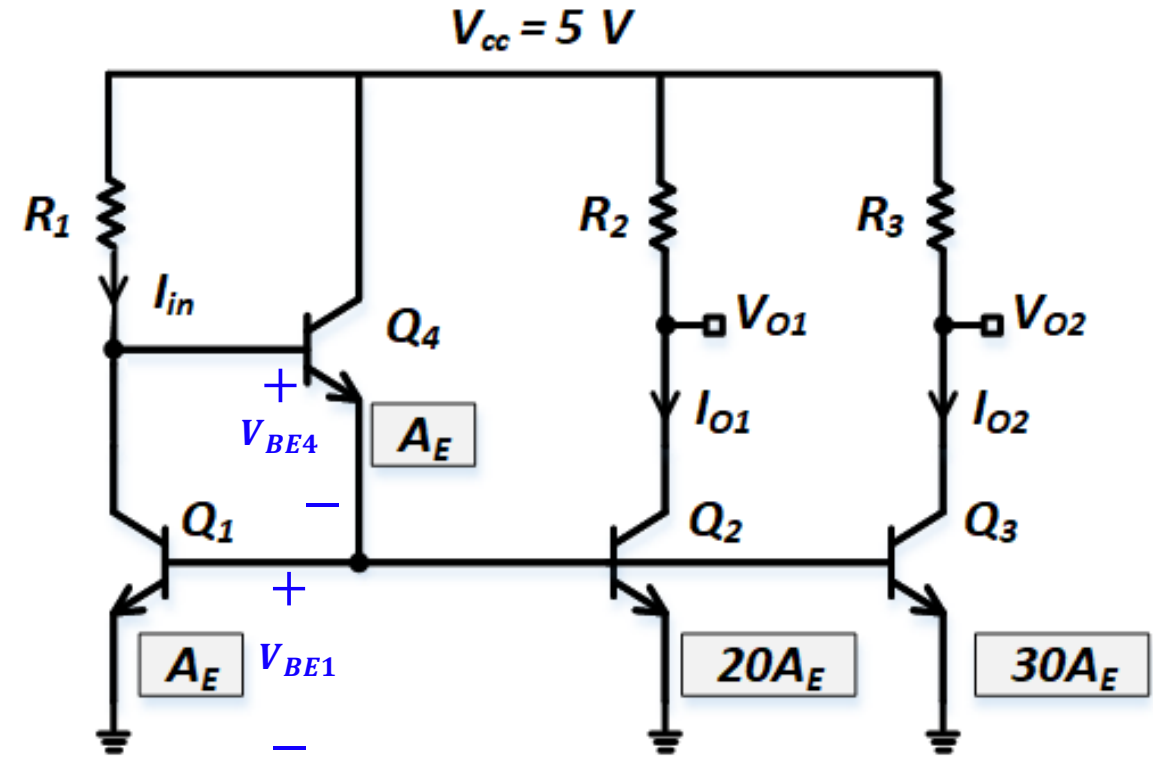
# Example #1 (1/3)

➤ If  $\beta = 120$ ,  $R_1 = 3.6\text{ K}\Omega$ ,  $R_2 = R_3 = 200\text{ }\Omega$ , find  $V_{o1}$  and  $V_{o2}$ .

❑ Q3 operates in saturation region

❑  $V_{o2} = V_{CE3} \approx 0.2\text{ V}$

❑  $I_{o2} = \frac{V_{CC} - V_{CE3}}{R_3} = \frac{5\text{ V} - 0.2\text{ V}}{200\text{ }\Omega} = 24\text{ mA}$



# References

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- Gray, Paul R., et al, “*Analysis and Design of Analog Integrated Circuits*”, 5<sup>th</sup> edition, John Wiley & Sons, 2009.
- Behzad Razavi, “*Fundamentals of Microelectronics*”, 2<sup>th</sup> edition, Wiley, 2014.
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□ Thanks